

Supplementary Information

Two-Stage Graph Estimation for Complex Motifs

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Appendix A Notation

The notation involved in the mathematical deductions of the algorithm and the estimation strategy are presented in Table.A.I.

Table A.I: Notation involved in the mathematical deductions.

Symbol	Definition	Meaning
$G = (V, E)$	Ordered pair of graph	An undirected or directed i-mode network.
n	Vertex counts	The number of vertexes in graph G .
A	Adjacency matrix	Adjacency matrix $A \in \{0, 1\}^{n \times n}$ corresponding to G
S	Subgraph	A subgraph with adjacency matrix $A_S \in \{0, 1\}^{k \times k}$
k	Vertex counts	The number of vertexes in subgraph S .
H	Induced subgraph	A subgraph with adjacency matrix $A_H \in \{0, 1\}^{k \times k}$ $k \times k$ while induced in the graph G .

Symbol	Definition	Meaning
ϕ	Labeled induced embedding	An injection from $[k]$ to V satisfying induced restrictions.
$I_H(G; \phi)$	Induced embedding indicator	Equal to 1 if ϕ satisfies all edge and non-edge constraints of H , and 0 otherwise.
$L_H(G)$	Labeled count	The number of all valid labeled induced embeddings.
$\text{Aut}(H)$	Automorphism	The automorphism group of H .
a_H	Automorphism size	$a_H = \text{Aut}(H) $.
$N_H(G)$	Unlabeled count	Size of the quotient set of the labeled embeddings by the automorphism group, i.e., $N_H = L_H/a_H$.
w_E	Edge-constraint weight	$w_E = -\log(\hat{p})$.
w_0	Non-edge-constraint weight	$w_0 = -\log(1 - \hat{p})$.
$\mathfrak{N}(u)$	Global constraint score	Total edge and non-edge constraint score of target-subgraph vertex u .
$\mathfrak{C}(u \mid \mathcal{S})$	Conditional constraint score	Constraint score of vertex u relative to the remaining target-subgraph vertices in $\mathcal{S}$.

Symbol	Definition	Meaning
$\mathbf{o} = (o_1, \dots, o_k)$	Ordering	Rare-constraint ordering of vertexes.
$V_{\neq}$	Remaining host-vertex set	Set of host-graph vertices not yet used by the current sampling path.
$N(x), \overline{N}(x)$	Neighbor and non-neighbor sets	Neighbor and non-neighbor sets of host vertex x , respectively.
C_t	Candidate set at step t	The set of unused host vertices satisfying the induced constraints relative to the already placed vertices.
x_t	Vertex drawn at step t	Drawn uniformly from C_t .
$g_{\mathbf{o}}(\phi)$	Proposal probability	Probability that ISS samples the complete embedding ϕ under ordering $\mathbf{o}$.
q_{ϕ}	Target contribution	Indicator or probability contribution of embedding ϕ to the target subgraph count.
W_b	Path weight in replication b	For a successful path, $W_b = \prod_t C_t $; for a failed path, $W_b = 0$.
B	Sampling times	Number of observed network sampling replications, used to estimate $\theta = N_H(G)$.

Symbol	Definition	Meaning
$\widehat{L}_H, \widehat{N}_H$	Count estimators	Estimators for labeled and unlabeled subgraph H count in observed graph G .
σ_H^2	Path-weight variance	Variance of the ISS path weight W_b for estimating the target subgraph count.
s_W^2	Sample variance of weights	Sample variance of the independent ISS path weights $\{W_b\}_{b=1}^B$.
ESS	Effective sample size	Diagnostic measuring the effective number of non-degenerate importance weights.
ESP	Effective sample proportion	Normalized effective sample size, defined as ESS/B .
CV_W	Coefficient of variation	Coefficient of variation of the ISS path weights.
$\rho_{\max}$	Maximum weight share	Largest normalized contribution of a single ISS weight.
π_0, π_+	Zero and positive weight proportions	Proportions of zero and strictly positive ISS weights, respectively.
$CI_{\text{Wald}}^{1-\alpha}$	Wald confidence interval	Standard Wald-type confidence interval for the ISS estimator.

Symbol	Definition	Meaning
$CI_{\log W}^{1-\alpha}$	Log-scale Wald confidence interval	Auxiliary confidence interval obtained from a log-scale approximation.
$\hat{F}_B$	Empirical weight distribution	Empirical distribution of the sampled ISS weights $\{W_b\}_{b=1}^B$.
$\mathcal{Q}_\gamma^*$	Bootstrap quantile	Empirical γ -quantile of the nonparametric bootstrap distribution.
$CI_{\text{boot}}^{1-\alpha}$	Bootstrap confidence interval	Nonparametric bootstrap confidence interval for the ISS estimator.
$\mathcal{G}_0$	Null model	The null model generated from graph G obeying specific rules.
$\mathcal{P}_0$	Probability matrix	The generation probability of each potential edge under the null model.
G^*	Null-model random network	A random network generated from $\mathcal{P}$.
$N_H(G^*)$	Real unlabeled count	Real unlabeled count of subgraph H in a specific random network G^* generated from the null model.

Symbol	Definition	Meaning
μ_0	Null-model expectation	Expected target subgraph count under the population null model $\mathcal{P}_0$.
R	Sampling times	Number of null-expectation sampling replications, used to estimate μ_H .
$g_t(x)$	Incremental probability	The product of edge and non-edge probabilities contributed by candidate x relative to the already placed vertices.
Z_t	Normalizing constant at step t	$Z_t = \sum_{x \notin \text{used}} g_t(x)$.
W_{0r}	Path weight in replication r	$W_{0r} = \prod_t Z_t$.
θ	Real count	Real count of subgraph H in graph G , i.e., $\theta = N_H(G)$.
$\hat{\theta}_B$	Estimated count	The unlabeled count of subgraph H in graph G , estimated by the first-stage ISS procedure.
$\hat{\mu}_R$	Estimated expectation	The unlabeled count expectation of subgraph H in null model, estimated by the second-stage ISS procedure.

Symbol	Definition	Meaning
s_{θ}^2, s_{μ}^2	Stage-specific sample variances	Sample variances of the observed-stage and null-stage ISS weights.
$\sigma_{\theta}^2, \sigma_{\mu}^2$	Stage-specific weight variances	Population variances of the observed-stage and null-stage ISS path weights.
β	Log enrichment coefficient	$\beta = \log(\theta/\mu)$.
$\hat{\beta}$	Log enrichment estimator	$\hat{\beta} = \log(\hat{\theta}_B/\hat{\mu}_R)$.
τ_0^2	Endogenous null-model variance	$\tau_0^2 = \mathbb{V}[N_H(G^*)]$.
Z_{β}	Motif test statistic	Standardized statistic for testing log-enrichment β .
Z_{β}^*	Calibrated test statistic	Studentized statistic used in random-network calibration.
p -value	Significance probability	Computed according to a right-tail alternative.
K	Number of calibration random networks	Used for a small number of random-network calibration draws.
β^*	Real Log enrichment	Real Log enrichment of a specific random-network generated from the null model.

Symbol	Definition	Meaning
$\hat{\beta}_k^*$	Log enrichment estimator	Estimator for log enrichment of a specific random-network generated from the null model, in k -th sampling.
$X_{k,n}$	k -clique count	Number of k -cliques in the random graph G_n .
e_k	Clique edge number	Number of edges in a k -clique, $e_k = k(k - 1)/2$.
I_S	Clique indicator	Indicator that vertex set S realizes a k -clique.
s	Overlap size	Number of shared vertices between two candidate k -cliques.
$\text{PA}(\Lambda_n, a_n)$	Pólya–Aeppli approximation	Compound Poisson approximation with intensity Λ_n and clumping parameter a_n .
D_n	Dispersion index	Relative variance measure used to characterize departure from the Poisson approximation.
w_{ni}	Chung–Lu expected degree	Expected degree parameter of vertex i in the n th Chung–Lu graph.
W_n	Total Chung–Lu weight	Total expected degree, $W_n = \sum_i w_{ni}$.

Symbol	Definition	Meaning
$A_{r,n}$	Degree-moment sum	Moment sum of Chung–Lu expected degrees, $A_{r,n} = \sum_i w_{ni}^r$.
$q_{S,T}$	Joint realization probability	Probability that two vertex sets S and T jointly realize the required clique constraints.
$\Omega_{k,s,n}$	Overlap contribution	Aggregate contribution from pairs of k -cliques sharing s vertices.
ω_β^2	Calibration variance	Variance of the random-network calibration distribution of log-enrichment.
$\hat{\omega}_\beta^2$	Estimated calibration variance	Sample estimate of ω_β^2 from calibration replications.
$\bar{\beta}^*$	Mean calibrated log-enrichment	Average log-enrichment across calibration random networks.
G_k^*	Calibration random graph	The k th random graph generated from the null model in calibration.
$\theta_k^*, \hat{\theta}_k^*$	Calibration count and estimator	True and estimated target subgraph count in calibration graph G_k^* .

Symbol	Definition	Meaning
$p_{K,\text{cal}}^+, p_{\text{cal}}^+$	Calibrated right-tail probabilities	Finite- K and population calibrated right-tail probabilities.
T_{obs}, T_r	Calibration statistics	Count, log, or excess statistics for the observed network and the r -th random network.

Appendix B Definition

This section provides the definitions that are not presented in detail in the main text because of space constraints. In particular, this section discusses the corresponding simple directed case.

B.1 Basic Definition of Graph

Formally, a 1-mode undirected network is usually represented as a simple undirected graph in graph theory, denoted by G , which can be defined as an ordered pair $G = (V, E)$.

Definition 1 (Simple undirected graph). Suppose that: (1) V denotes the vertex set satisfying $|V| = n$; and (2) E denotes the edge set consisting of unordered pairs of vertexes satisfying $E \subseteq \{\{i, j\} \in V^2 : i \neq j\}$. Then the ordered pair $G = (V, E)$ is called a simple undirected graph with n vertexes.

Correspondingly, a 1-mode directed network can be defined as an ordered pair $G = (V, E)$, with the edge set E being ordered pairs.

Definition 2 (Simple directed graph). For a vertex set V satisfying $|V| = n$, and an edge set E consisting of ordered pairs of vertexes satisfying $E \subseteq \{(i, j) \in V^2 : i \neq j\}$. Then the ordered pair $G = (V, E)$ is called a simple directed graph with n vertexes.

The adjacency matrix of G can be denoted by $A \in \{0, 1\}^{n \times n}$, satisfying: (1) A is symmetric, i.e., $A_{ij} = A_{ji}$; and (2) all diagonal entries are zero (irrespective of loop), i.e., $A_{ii} = 0$. In the directed case, E is the edge set consisting of ordered pairs of vertexes satisfying $E \subseteq \{(i, j) \in V^2 : i \neq j\}$. Then, the adjacency matrix A : (1) is not required to be symmetric; and (2) has $A_{ij} = 1$ when there exists a directed edge e_{ij} from vertex i to vertex j , with the direction denoted here by $i \rightarrow j$. On the basis of the graph, we may define a subgraph, denoted by S , which is likewise represented as an ordered pair $S = (V_S, E_S)$.

Definition 3 (Subgraph). For any vertex set $V_S \subseteq V$ and any edge set $E_S \subseteq E$ satisfying $E_S \subseteq \{(i, j) \in V_S^2 : i \neq j\}$, the ordered pair $S = (V_S, E_S)$ is called a subgraph of the graph $G = (V, E)$.

As in the case of the graph, the adjacency matrix of a subgraph S with k vertexes is denoted by $A_S \in \{0, 1\}^{k \times k}$, which also satisfies the requirements of symmetry and zero diagonal entries. In the directed case, S is a simple directed subgraph, and its adjacency matrix satisfies the usual requirements for the adjacency matrix of a directed graph with k vertexes.

Based on the definition of subgraph, we now define the “connected subgraph”.

Definition 4 (Connected Subgraph). For $\forall u, v \in V_S$ where $u \neq v$, $\exists m \in \mathbb{N}$ and $\exists (v_0, \dots, v_m) \in V_S^{m+1}$, satisfying $\{v_{\ell-1}, v_\ell\} \in E_S$, $\ell = 1, \dots, m$, where $v_0 = u$ and $v_m = v$. Then, $G_S = (V_S, E_S)$ is a connected subgraph. In the directed case, the unordered pair $\{v_{\ell-1}, v_\ell\} \in E_S$ should be ordered, i.e., $(v_{\ell-1}, v_\ell) \in E_S$.

Intuitively, the connectedness of a connected subgraph means that every vertex in the subgraph can be reached through at least 1 path, whose edges all belong to the original graph. Equivalently, a connected subgraph does not contain two or more completely separate components.

Defs. 1–4 provide the basic concepts used in complex network analysis. In most cases, these 3 definitions are also sufficient to give a definition for “motif”, which is usually referred to as a “non-induced” type. In complex networks, however, non-inducedness imposes only a relatively weak set of edge constraints. To estimate subgraph counts effectively in a network, we also need to introduce several more restrictive definitions. The first is the “induced subgraph”, which differs from a general subgraph.

Definition 5 (Induced subgraph). Given a graph $G = (V_G, E_G)$ and a vertex subset $S \subseteq V_G$, if a subgraph H satisfies $H = (V_H, E_H) = (S, E_G \cap \{\{i, j\} : i, j \in S, i \neq j\})$, then we denote H as $G[S]$, and H is called the induced subgraph of G induced by S . In the directed case, the edge set E_G of subgraph H should satisfy $E_G \cap \{(i, j) : i, j \in S, i \neq j\}$.

Intuitively, “induction” emphasizes that the edge structure of the subgraph is completely determined by the connections among the selected vertexes in the original network. It is worth noting that the key difference between an induced subgraph and a subgraph is that, an induced subgraph must retain all edges contained in G among the selected vertexes, whereas a subgraph need not contain all such edges.

Then, based on Definition 1–5, we can give a more strict definition for a subgraph, namely, “connected induced subgraph”.

Definition 6 (Connected induced subgraph). Suppose that: (1) the graph H is a subgraph of G ; (2) the graph H is connected; and (3) the graph H is induced. Then, the graph H is called the connected induced subgraph (CIS) of graph G .

With the concepts of graph and subgraph defined by Defs. 1–6, we can now define a “motif”. In fact, there is currently no fully settled consensus on the precise definition of a motif. Some studies define a motif as a subgraph with a particular connection pattern, whereas other studies define a motif as a subgraph whose frequency is no smaller than that in random networks generated under a specified rule. Here we provide a more explicit statistical definition:

Definition 7 (Motif). Given a graph G , a target connected induced subgraph H , a random-network population distribution $\mathcal{G}_0$, and a significance level α , if the right-tail probability satisfies $\mathbb{P}\{N_H(G^*) \geq N_H(G) \mid G^* \sim \mathcal{G}_0\} \leq \alpha$, then H is called a (genuine) motif in G .

Intuitively, this definition states that a motif is a subgraph occurring significantly more frequent in the targeted network than in random networks generated under a specified rule.

B.2 Definition for Sampling Procedure

For the purpose of counting the subgraph in a given graph, we should investigate the “embedding” status of subgraph in the given graph. Thus, we now define the mapping of an induced subgraph into the original graph G :

Definition 8 (Induced embedding). Let an injection $\phi : V_H \hookrightarrow V_G$ satisfy, for $\forall a, b \in V_H$, the constraint $A_H(a, b) = A_G(\phi(a), \phi(b))$ holds. Then ϕ is called an “induced embedding” of the target subgraph H in the graph G .

The constraint condition is hereafter referred to as the “induced constraint”. The matrices A_H and A_G are the adjacency representations of the subgraph H and the graph G , respectively.

Definition 9 (Labeled induced embedding). Further, fix a labeling $S = (s_1, s_2, \dots, s_k)$ of the vertex set of the target subgraph H , and also let ϕ be an injection of vertex set V_H into the vertex set V_G , denoted as $\phi : V_H \hookrightarrow V_G$. When: (1) the injection ϕ can be represented by ordered tuples $(\phi(s_1), \dots, \phi(s_k)) = (v_1, \dots, v_k)$, where $v_1, \dots, v_k$ are distinct vertices in G ; and (2) for $\forall a, b \in S$, the induced constraint $A_H(a, b) = A_G(\phi(a), \phi(b))$ holds, then ϕ is called a “labeled induced embedding” of the target subgraph H in the graph G .

On the basis of Def.9, let the set of all labeled induced embeddings be denoted by $\text{Emb}_{\text{ind}}(H, G) = \{\phi : V_H \hookrightarrow V_G : G[\phi(V_H)] \cong_{\text{ind}} H\}$, i.e., the set of all injections under

which the subgraph induced by $\phi(V_H)$ is isomorphic to H . Then, by defining the indicator function $I_H(G; \phi) = \mathbb{1}\{A_G(\phi(a), \phi(b)) = H_{ab} \mid \forall (a, b) \in V_H^2\}$, the total number of labeled induced embeddings of the target subgraph H is:

$$L_H(G) = \sum_{\phi \in \text{Inj}([k], V_G)} I_H(G; \phi) = |\text{Emb}_{\text{ind}}(H, G)| \quad (\text{B.1})$$

where $\text{Inj}([k], V_G)$ denotes the set of all injections from the labeled vertexes $\{s_1, s_2, \dots, s_k\}$ to V_G . Estimating the size of the target subgraph requires not only estimating the total number of labeled induced embeddings, but also using information on the automorphisms of the target subgraph, thus, we have:

Definition 10 (Automorphism). Let a bijection $\pi : V_H \xrightarrow{\sim} V_H$ satisfy: (1) in the undirected case, $\{u, v\} \in E_H \iff \{\pi(u), \pi(v)\} \in E_H, \forall u, v \in V_H$; or (2) in the directed case, the condition is $(u, v) \in E_H \iff (\pi(u), \pi(v)) \in E_H$. Then $\text{Aut}(H) = \{\pi : V_H \xrightarrow{\sim} V_H : \forall u, v \in V_H, u \neq v, A_H(u, v) = A_H(\pi(u), \pi(v))\}$ is called the automorphism group of H , and its automorphism number is $a_H = |\text{Aut}(H)|$.

On the basis of the definition of automorphism, we now define an equivalence relation:

Definition 11 (Equivalence class). If there exists $\pi \in \text{Aut}(H)$ such that $\psi = \phi \circ \pi$, then ϕ and ψ are said to be equivalent, denoted by $\phi \sim \psi$. The equivalence class represented by ϕ , which is also

the equivalence class represented by ψ , is an unlabeled induced embedding of the target subgraph H , denoted by $[\phi] = [\psi] = \{\phi \circ \pi : \pi \in \text{Aut}(H)\}$.

Therefore, the set of all unlabeled induced embeddings is the quotient set of the labeled embeddings by the automorphism group. The size of the target subgraph can then be written as:

$$N_H(G) = \left| \frac{\text{Emb}_{\text{ind}}(H, G)}{\text{Aut}(H)} \right| = \frac{L_H(G)}{a_H} \quad (\text{B.2})$$

Appendix C Stage I: Subgraph Estimation in Targeted Graph

This section provides the design of the first-stage estimation strategy not stated in detail in the main text because of space constraints, including the corresponding directed case, the proofs of the propositions, and the pipeline of the algorithm designed in the main text.

C.1 Design and Proof

The first step of the first-stage estimation is to set constraint weights. Given the target subgraph H and the density of the target network $\hat{p}_G = \binom{n}{2}^{-1}|E|$, we define the edge-constraint and non-edge-constraint weights as:

$$\begin{cases} w_E = -\log(\hat{p}_G) \\ w_0 = -\log(1 - \hat{p}_G) \end{cases} \quad (\text{C.3})$$

Intuitively, a sparser network means rarer edge constraints and a larger w_E , while a denser network means rarer non-edge constraints and a larger w_0 .

The second step is to specify the sampling rule for the first and subsequent subgraph vertexes.

For a subgraph vertex u , we define its global constraint score and conditional constraint score as:

$$\begin{cases} \mathfrak{N}(u) = \sum_{v \neq u} [H_{uv}w_E + (1 - H_{uv})w_0] \\ \mathfrak{C}(u \mid \mathcal{S}) = \sum_{s \in \mathcal{S}} [H_{us}w_E + (1 - H_{us})w_0] \end{cases} \quad (\text{C.4})$$

where $\mathcal{S}$ denotes the set of vertexes that have not yet been sampled. The first sampling step selects the vertex with the highest global constraint score $\mathfrak{N}(u)$, and each subsequent step selects, from the unsampled set $\mathcal{S}$, the vertex with the highest conditional constraint score $\mathfrak{C}(u \mid \mathcal{S})$. Intuitively, imposing stronger constraints earlier shrinks subsequent candidate sets more quickly, thereby reducing invalid sampling paths and sampling variance.

The third step is to construct candidate vertex sets. Let the constraint ordering of the target subgraph be $\mathbf{o} = (o_1, o_2, \dots, o_k)$. At step t , suppose that the selected target-network vertexes are $\mathbf{x} = (x_1, x_2, \dots, x_{t-1})$, where x_s corresponds to subgraph vertex o_s . The candidate set C_t consists of all vertexes that have not yet been sampled and that satisfy the induced constraints relative to the already sampled vertexes:

$$C_t(\phi) = \{v \in V_{\neq} : G[v, x_j] = H[o_t, o_j], \forall j < t\}. \quad (\text{C.5})$$

where $V_{\neq} = V_{\setminus (x_1, x_2, \dots, x_{t-1})}$ denotes the vertex set after removing vertexes $(x_1, x_2, \dots, x_{t-1})$. While

in the undirected case, let $N(x_s)$ denote the neighbor vertex set of x_s , and let $\overline{N}(x_s)$ denote the non-neighbor vertex set of x_s . In implementation, $N(x_s)$ and $\overline{N}(x_s)$ are stored as logical bitsets, and the candidate set $C_t(\phi)$ can be obtained by successive intersections:

$$C_t = V_{\neq} \cap \left(\bigcap_{s < t: H_{o_t, o_s} = 1} N(x_s) \right) \cap \left(\bigcap_{s < t: H_{o_t, o_s} = 0} \overline{N}(x_s) \right) \quad (\text{C.6})$$

And in the directed case, two directional constraints exist between each already sampled vertex and the current candidate vertex. If the current candidate vertex x corresponds to subgraph vertex o_t , and the sampled vertex x_s corresponds to o_s , then both directions ($x \rightsquigarrow x_s$ and $x_s \rightsquigarrow x$) must be checked. The candidate set can be expressed as:

$$C_t = V_{\neq} \cap \left(\bigcap_{s < t} C_{t,s}^{\text{out}} \right) \cap \left(\bigcap_{s < t} C_{t,s}^{\text{in}} \right) \quad (\text{C.7})$$

where $C_{t,s}^{\text{out}} = N^-(x_s)$ if $H_{o_t, o_s} = 1$, or becomes non-adjacency set otherwise; $C_{t,s}^{\text{in}} = N^+(x_s)$ if $H_{o_s, o_t} = 1$, or becomes non-adjacency set otherwise.

At any step of a sequential sampling process, if $C_t = \emptyset$, the path weight of that entire sampling will be 0; otherwise x_t is sampled uniformly from C_t . The weight W of a successful sampling path is

the product of $|C_t|$ over all steps, i.e.:

$$W = \prod_{t=1}^k |C_t| \quad (\text{C.8})$$

For the conditional sampling probability at step t being $1/|C_t|$, the sampling probability of a successful path is $q(x_1, x_2, \dots, x_k) = \prod_t |C_t|^{-1} = W^{-1}$. Hence the path weight W is exactly the inverse probability of the sampled path.

Let $\{W_1, W_2, \dots, W_B\}$ be the path weights obtained from B independent sampling processes. We define the labeled and unlabeled subgraph-count estimators for the $\mathbf{r}$ -mode network as:

$$\begin{cases} \hat{L}_H(G) = \frac{1}{B} \sum_{b=1}^B W_b \\ \hat{N}_H(G) = \frac{\hat{L}_H(G)}{a_H} \end{cases} \quad (\text{C.9})$$

and given the target network and the target subgraph, we have the following proposition:

Proposition 1 (Unbiasedness). Both the labeled and unlabeled ISS estimators are unbiased, i.e., $\mathbb{E}[\hat{L}_H] = L_H$ and $\mathbb{E}[\hat{N}_H] = N_H$.

Proof. Given an ordering $\mathbf{o} = (o_1, o_2, \dots, o_k)$, each valid labeled induced embedding ϕ corresponds to a unique successful sampling path $\mathbf{x} = (x_1, x_2, \dots, x_k)$, where $x_t = \phi(o_t)$. Along this path, the size of the candidate set at step t is $|C_t(\phi)|$, and the path probability is $q(\phi) = \prod_t \frac{1}{|C_t(\phi)|}$. The algorithm assigns such a path with the weight $W(\phi) = \prod_t |C_t(\phi)| = \frac{1}{q(\phi)}$, and invalid paths with the weight zero. Therefore,

$$\mathbb{E}[W] = \sum_{\text{valid } \phi} q(\phi)W(\phi) + \sum_{\text{invalid } \phi^*} q(\phi^*) \cdot 0 = L_H(G) \quad (\text{C.10})$$

and thus,

$$\mathbb{E}[\widehat{L}_H(G)] = \frac{1}{B} \sum_{b=1}^B \mathbb{E}[W_b] = L_H(G) \quad (\text{C.11})$$

Furthermore, since the number of unlabeled induced subgraph can be estimated by:

$$N_H(G) = \frac{L_H(G)}{a_H} \quad (\text{C.12})$$

and we then have:

$$\mathbb{E}[\widehat{N}_H(G)] = \frac{\mathbb{E}[\widehat{L}_H(G)]}{a_H} = \frac{L_H(G)}{a_H} = N_H \quad (\text{C.13})$$

Q. E. D.

□

Now, consider the sampling asymptotic properties conditional on given G and H while $B \rightarrow \infty$, we show that:

Proposition 2 (Consistency). Since W_b are i.i.d. and bounded in a finite network, $\hat{N}_H(G)$ is a consistent estimator of $N_H(G)$. The sample variance $s_W^2 = \frac{1}{B-1} \sum_b \left(W_b - \hat{L}_H\right)^2$ is also a consistent estimator of σ_H^2 .

Proof. First, we will show that, $\hat{N}_H(G)$ is a consistent estimator of $N_H(G)$. Conditional on G and H , the candidate set at each step has size at most n . Therefore, at least we have $0 \leq W_b \leq n^k$, so that $\mathbb{E}[W_b] < \infty$, and W_b is bounded in a finite network. Since W_b are i.i.d., the Strong Law of Large Numbers (SLLN) gives:

$$\frac{1}{B} \sum_{b=1}^B W_b \xrightarrow{a.s.} \mathbb{E}(W_b) \quad (\text{C.14})$$

that is, $\hat{L}_H(G) \xrightarrow{a.s.} L_H(G)$. And dividing both sides by the constant a_H , we obtain:

$$\hat{N}_H(G) = \frac{\hat{L}_H(G)}{a_H} \rightarrow \frac{L_H(G)}{a_H} = N_H(G) \quad (\text{C.15})$$

thus, we have: $\widehat{N}_H(G) \xrightarrow{a.s.} N_H(G)$.

Second, we will show that, $s_W^2 = \frac{1}{B-1} \sum_b \left(W_b - \widehat{L}_H\right)^2$ is also a consistent estimator of σ_H^2 . For $\sigma_H^2 = \mathbb{V}[W_b]$ and $\widehat{L}_H(G) = \frac{1}{B} \sum_{b=1}^B W_b$, the sample variance can be rewritten as:

$$s_W^2 = \frac{B}{B-1} \left[\frac{1}{B} \sum_{b=1}^B W_b^2 - \left(\frac{1}{B} \sum_{b=1}^B W_b \right)^2 \right] \quad (\text{C.16})$$

Since W_b is bounded in a finite network, i.e., $\mathbb{E}[W_b^2] < \infty$. SLLN gives:

$$\begin{cases} \frac{1}{B} \sum_{b=1}^B W_b \xrightarrow{a.s.} \mathbb{E}[W_b] \\ \frac{1}{B} \sum_{b=1}^B W_b^2 \xrightarrow{a.s.} \mathbb{E}[W_b^2] \end{cases} \quad (\text{C.17})$$

therefore,

$$\left(\frac{1}{B} \sum_{b=1}^B W_b \right)^2 \xrightarrow{a.s.} (\mathbb{E}[W_b])^2 \quad (\text{C.18})$$

Since $\lim_{B \rightarrow \infty} \frac{B}{B-1} = 1$, we have:

$$s_W^2 \xrightarrow{a.s.} \mathbb{E}[W_b^2] - (\mathbb{E}[W_b])^2 = \mathbb{V}[W_b] = \sigma_H^2. \quad (\text{C.19})$$

Thus, s_W^2 is a consistent estimator of σ_H^2 . Hence,

$$\begin{cases} \mathbb{V}[\widehat{L}_H(G)] = \frac{\sigma_H^2}{B} \\ \mathbb{V}[\widehat{N}_H(G)] = \frac{\sigma_H^2}{Ba_H^2} \end{cases} \quad (\text{C.20})$$

therefore, the Monte Carlo standard error of the unlabeled count estimator is

$$\text{SE}[\widehat{N}_H(G)] = \frac{s_W}{a_H \sqrt{B}}. \quad (\text{C.21})$$

Q. E. D. □

C.2 Confidence Intervals

As mentioned above, under standard large-sample conditions, if W_b has a finite second moment and there is no case in which a single weight dominates the summative one, then the CLT gives

$\sqrt{B} \left[\hat{N}_H(G) - N_H(G) \right] \stackrel{\text{approx}}{\sim} \mathcal{N}(0, \sigma_H^2/a_H^2)$. By constructing the z -statistic, the above test can provide us with a typical Wald-type confidence interval (CI). At the $1 - \alpha$ level, a typical Wald interval can be:

$$\text{CI}_{Wald}^{1-\alpha}(\theta) = \left[\hat{\theta} - z_{1-\alpha/2} \frac{s_W}{a_H \sqrt{B}}, \quad \hat{\theta} + z_{1-\alpha/2} \frac{s_W}{a_H \sqrt{B}} \right] \quad (\text{C.22})$$

The Wald-type CI, directly corresponding to the CLT approximation of the estimator, is therefore easy to interpret. Moreover, in cases of non-rare subgraphs, high effective sample size, or a weight distribution being not dominated by extreme paths, such a CI usually has good finite-sample performance.

C.2.1 Sample Diagnostics

However, for the target subgraph count being non-negative, and since it is zero-inflated or highly skewed in distribution, the Wald CIs may have under-coverage and give a negative lower bound. It is worth noting that the reliability of ISS path sampling in finite samples depends not only on the number of samples B , but also the asymmetric nature or skewness of the weight distribution. For such a reason, here, we first construct several sample diagnostics for path sampling, thereby helping us judge whether the typical Wald-type CIs can provide good finite-sample performance.

First, we define the effective sample proportion (ESP), namely $ESP = \frac{ESS}{B} = \frac{(\sum_{b=1}^B W_b)^2}{B \sum_{b=1}^B W_b^2}$;

where $ESS = \frac{(\sum_{b=1}^B W_b)^2}{\sum_{b=1}^B W_b^2}$ denotes the effective sample size (ESS) of the ISS path-sampling weights. Globally identical (i.e., uniform contribution) within the ISS path-sampling weights means $ESP = 1$; while, once the ISS weights are highly heterogeneous (i.e., a few extremely large weights play a dominant role), then ESP will approach 0. Thus, ESP, in fact, measures the proportion of effective information in the weight distribution relative to the case of uniform contribution.

Second, we define the coefficient of variation (CV) of the weights as $CV_W = \frac{s_W}{\bar{W}}$; where $\bar{W} = \frac{1}{B} \sum_{b=1}^B W_b$. The larger CV_W is, the more dispersed the weights are relative to their mean, and the more likely the estimator is to be affected by high-weight paths. Under a fixed number of samples, a larger CV_W usually implies a larger Monte Carlo standard error.

Third, we define the maximum weight share as $\rho_{\max} = \frac{\max_{1 \leq b \leq B} W_b}{\sum_{b=1}^B W_b}$. The larger $\rho_{\max}$ is, the more it indicates that a single sampling path contributes too much to the total estimator. In such a case, even if the standard error is finite, the estimation result may still be highly sensitive to a few paths.

Finally, within one round of ISS given a fixed graph, if a sampling path cannot satisfy the edge or non-edge constraints in subsequent steps, it may obtain a zero weight. Therefore, we define the zero-weight proportion $\pi_0 = \frac{1}{B} \sum_{b=1}^B \mathbf{1}\{W_b = 0\}$ and the positive-weight proportion $\pi_+ = \frac{1}{B} \sum_{b=1}^B \mathbf{1}\{W_b > 0\}$. In subgraph estimation for a simple fixed graph, π_+ is closely related

to the sampling success rate. If π_+ is certainly low, then the estimator contains a large number of zero weights and a small number of nonzero weights, and the ordinary normal approximation may converge more slowly.

C.2.2 Auxiliary Confidence Intervals

To avoid slow convergence of the CLT approximation, leading to under-coverage of typical Wald-type CIs, we first consider to calculate the CI on the log-scale with the log-transformed point estimate and then back-transform by exponentiating the limits of the CI. Since $\hat{\theta} \geq 0$, let $h(\hat{\theta}) = \log \hat{\theta}$; and at this time, $h'(\hat{\theta}) = \frac{1}{\hat{\theta}}$. Applying the delta method to $\text{SE} \left[h \left(\hat{\theta} \right) \right]$, we then have:

$$\text{SE} \left[\log \left(\hat{\theta} \right) \right] = \text{SE} \left[\hat{\theta} \right] \cdot \left| \frac{\partial h \left(\hat{\theta} \right)}{\partial \hat{\theta}} \right| = \frac{\text{SE} \left[\hat{\theta} \right]}{\hat{\theta}} = \frac{s_W}{a_H \sqrt{B\hat{\theta}}} \quad (\text{C.23})$$

Thus, we can give the Wald CI for the log-transformed target subgraph count, by $\log(\hat{\theta}) \pm z_{1-\alpha/2} \frac{s_W}{a_H \sqrt{B\hat{\theta}}}$. Finally, via back-transformation based on exponentiation, we can obtain the log-scale

Wald CI:

$$\text{CI}_{\log W}^{1-\alpha}(\theta) = \left[\frac{\hat{\theta}}{\exp \left\{ z_{1-\alpha/2} \frac{s_W}{a_H \sqrt{B\hat{\theta}}} \right\}}, \hat{\theta} \cdot \exp \left\{ z_{1-\alpha/2} \frac{s_W}{a_H \sqrt{B\hat{\theta}}} \right\} \right] \quad (\text{C.24})$$

The log-scale Wald CI calculated based on the log-transformed point estimate and back-transformation by exponentiation still relies on large-sample approximation. If the target subgraph is highly skewed, the log-scale Wald CI will be still incapable to eliminate the problem caused by insufficient proposal coverage. In this regard, in order to better reflect distributional skewness and asymmetry, we also consider constructing a non-parametric bootstrap CI based on quantiles. In brief, this method performs resampling with replacement on the ISS path-sampling weight sample, and approximates the finite-sample sampling distribution of the ISS estimator using the empirical weight distribution.

For a given ISS path-sampling weight sample $\{W_1, \dots, W_B\}$, define the empirical distribution $\hat{F}_B = \frac{1}{B} \sum_{b=1}^B \delta_{W_b}$, where δ_{W_b} denotes the Dirac measure at the point W_b . Sampling with replacement from $\hat{F}_B$, the weights from the m -th bootstrap resampling satisfy $W_1^{*(m)}, \dots, W_B^{*(m)} \stackrel{i.i.d.}{\sim} \hat{F}_B$; equivalently, let $t_1^{*(m)}, \dots, t_B^{*(m)} \stackrel{i.i.d.}{\sim} \text{Uniform}\{1, \dots, B\}$, then $W_b^{*(m)} = W_{t_b^{*(m)}}^{*}$. Thus, the m -th bootstrap estimator is $\hat{\theta}^{*(m)} = \frac{1}{B} \sum_{b=1}^B W_b^{*(m)}$. After repeating M times, we can obtain the boot-

strap distribution $\mathcal{L}^* \left(\hat{\theta}^{*(1)}, \dots, \hat{\theta}^{*(M)} \right)$. Let $\mathcal{Q}_\gamma^*$ denote the empirical γ quantile of $\mathcal{L}^*$. Then the quantile-based non-parametric bootstrap interval will be:

$$\text{CI}_{boot}^{1-\alpha}(\theta) = [\mathcal{Q}_{\alpha/2}^*, \mathcal{Q}_{1-\alpha/2}^*] \quad (\text{C.25})$$

C.3 Sensitivity Analysis for Vertex Ordering

The ordering $\mathbf{o}$ is essentially a design of the proposal for path sampling, and does not constitute any component of the estimand. Therefore, for any fixed admissible ordering $\mathbf{o}$, the ISS estimator remains unbiased, provided that the proposal $g_{\mathbf{o}}(\phi)$ assigns positive probability (i.e., $g_{\mathbf{o}}(\phi) > 0$) to every embedding q_ϕ with nonzero target contribution (i.e., $q_\phi > 0$).¹ Nonetheless, the ordering will still affect the Monte Carlo variance, for different orderings substantially affecting the weight distribution.

Here, we report sensitivity analyses for subgraph estimation based on different vertex orderings. We generate random networks with 50 vertices ($n = 50$) and density 0.1 ($\hat{p}_G = 0.1$) according to the Erdős–Rényi rule. The subgraph used in the sensitivity analyses is the “Stingray”. The vertex orderings considered include the sequential importance adopted by ISS, which is based on conditional

¹Here, under a given ordering $\mathbf{o}$, $g_{\mathbf{o}}(\phi)$ is the probability of ISS algorithm drawing a complete embedding ϕ , which can also be called the proposal (probability) for path sampling.

and global constraints; the reverse of this sequential importance; and 15 distinct random orderings. Together with the sample diagnostic statistics, Wald CIs, and log-scale Wald CIs mentioned above, we present the sensitivity analysis results in Figs.C.2 and C.3. The results show that subgraph estimates based on different vertex orderings are generally consistent—although the point estimates differ somewhat, the CIs of all estimates overlap substantially. Meanwhile, the sequential importance adopted by ISS yields the smallest Monte Carlo variance among all vertex orderings.

Table C.2: Sensitivity analyses based on different importance sequences.

	Order	Vertex	$\hat{N}$	SE	95% CI (lwr.)	95% CI (upr.)
1	ISS	$1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5$	126.288	2.592	121.207	131.369
2	reverse	$5 \rightarrow 4 \rightarrow 3 \rightarrow 2 \rightarrow 1$	139.258	11.882	115.969	162.546
3	random	$5 \rightarrow 4 \rightarrow 3 \rightarrow 1 \rightarrow 2$	138.807	10.760	117.718	159.895
4	random	$2 \rightarrow 5 \rightarrow 4 \rightarrow 1 \rightarrow 3$	128.808	4.499	119.991	137.626
5	random	$5 \rightarrow 4 \rightarrow 3 \rightarrow 1 \rightarrow 2$	140.180	10.868	118.879	161.482
6	random	$4 \rightarrow 1 \rightarrow 5 \rightarrow 3 \rightarrow 2$	129.567	2.986	123.715	135.419
7	random	$5 \rightarrow 4 \rightarrow 3 \rightarrow 2 \rightarrow 1$	113.603	10.493	93.038	134.169
8	random	$5 \rightarrow 1 \rightarrow 3 \rightarrow 4 \rightarrow 2$	124.876	3.249	118.508	131.243

	Order	Vertex	$\hat{N}$	SE	95% CI (lwr.)	95% CI (upr.)
9	random	$5 \rightarrow 1 \rightarrow 2 \rightarrow 4 \rightarrow 3$	127.957	2.632	122.799	133.115
10	random	$1 \rightarrow 5 \rightarrow 4 \rightarrow 3 \rightarrow 2$	129.493	4.122	121.414	137.572
11	random	$2 \rightarrow 5 \rightarrow 1 \rightarrow 3 \rightarrow 4$	125.227	3.336	118.688	131.766
12	random	$3 \rightarrow 5 \rightarrow 2 \rightarrow 1 \rightarrow 4$	128.905	3.907	121.247	136.563
13	random	$4 \rightarrow 2 \rightarrow 5 \rightarrow 1 \rightarrow 3$	124.073	4.089	116.059	132.087
14	random	$5 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 1$	128.775	8.447	112.218	145.332
15	random	$4 \rightarrow 3 \rightarrow 2 \rightarrow 5 \rightarrow 1$	107.488	13.676	80.683	134.293
16	random	$1 \rightarrow 4 \rightarrow 5 \rightarrow 3 \rightarrow 2$	119.740	3.651	112.584	126.897
17	random	$3 \rightarrow 4 \rightarrow 1 \rightarrow 5 \rightarrow 2$	126.362	5.109	116.349	136.375

Table C.3: Sensitivity analyses with alternative indicators based on different importance sequences.

	Order	Vertex	log-W. (lwr.)	log-W. (upr.)	π_+	ESP	CV	$\rho_{\max}$
1	ISS	$1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5$	121.308	131.473	0.035	0.023	6.492	0.001
2	reverse	$5 \rightarrow 4 \rightarrow 3 \rightarrow 2 \rightarrow 1$	117.812	164.607	0.001	0.001	26.982	0.013
3	random	$5 \rightarrow 4 \rightarrow 3 \rightarrow 1 \rightarrow 2$	119.242	161.582	0.002	0.002	24.513	0.008

	Order	Vertex	log-W. (lwr.)	log-W. (upr.)	π_+	ESP	CV	$\rho_{\max}$
4	random	$2 \rightarrow 5 \rightarrow 4 \rightarrow 1 \rightarrow 3$	120.286	137.935	0.013	0.008	11.045	0.004
5	random	$5 \rightarrow 4 \rightarrow 3 \rightarrow 1 \rightarrow 2$	120.419	163.186	0.002	0.002	24.517	0.011
6	random	$4 \rightarrow 1 \rightarrow 5 \rightarrow 3 \rightarrow 2$	123.845	135.553	0.026	0.018	7.287	0.002
7	random	$5 \rightarrow 4 \rightarrow 3 \rightarrow 2 \rightarrow 1$	94.792	136.148	0.001	0.001	29.208	0.016
8	random	$5 \rightarrow 1 \rightarrow 3 \rightarrow 4 \rightarrow 2$	118.668	131.408	0.023	0.015	8.227	0.002
9	random	$5 \rightarrow 1 \rightarrow 2 \rightarrow 4 \rightarrow 3$	122.902	133.220	0.035	0.023	6.503	0.001
10	random	$1 \rightarrow 5 \rightarrow 4 \rightarrow 3 \rightarrow 2$	121.661	137.830	0.015	0.010	10.067	0.002
11	random	$2 \rightarrow 5 \rightarrow 1 \rightarrow 3 \rightarrow 4$	118.856	131.940	0.021	0.014	8.425	0.002
12	random	$3 \rightarrow 5 \rightarrow 2 \rightarrow 1 \rightarrow 4$	121.470	136.796	0.014	0.011	9.586	0.003
13	random	$4 \rightarrow 2 \rightarrow 5 \rightarrow 1 \rightarrow 3$	116.313	132.351	0.011	0.009	10.421	0.004
14	random	$5 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 1$	113.239	146.443	0.008	0.002	20.744	0.013
15	random	$4 \rightarrow 3 \rightarrow 2 \rightarrow 5 \rightarrow 1$	83.764	137.932	0.001	0.001	40.236	0.023
16	random	$1 \rightarrow 4 \rightarrow 5 \rightarrow 3 \rightarrow 2$	112.793	127.115	0.017	0.011	9.643	0.002
17	random	$3 \rightarrow 4 \rightarrow 1 \rightarrow 5 \rightarrow 2$	116.736	136.782	0.007	0.006	12.785	0.002

C.4 Implementation

Here, we give the pipeline of implementing the ISS algorithm for estimating subgraph in a give network.

Algorithm C.1: ISS algorithm for targeted network.

```
1      Input:  $n$ -vertex network  $G$ ,  $k$ -vertex subgraph  $H$ , sample times
2       $\hookrightarrow B$ .
3      Output: count of subgraph  $H$ .
4      1. Normalize  $A$  and  $H$  as simple undirected zero-diagonal
5       $\hookrightarrow$  matrices.
6      2. Compute graph density  $\hat{p}$  of targeted network and rare-
7       $\hookrightarrow$  constraint order  $\mathbf{o} = (o_1, o_2, \dots, o_k)$ .
8      3. Build bitsets for adjacency  $N(v)$ , non-adjacency  $\bar{N}(v)$ , and
9       $\hookrightarrow$  all vertices.
10     4. Conduct sampling via:
11         for  $b = 1, 2, \dots, B$ :
12          $X \leftarrow \text{empty}$ 
13          $W_b \leftarrow 1$ 
14         for  $t = 1, 2, \dots, k$ :
15          $C_t \leftarrow$  unused vertices satisfying all constraints to  $o_1, o_2,$ 
16          $\hookrightarrow \dots, o_{t-1}$ .
```

```

13   if  $C_t$  is empty:
14      $W_b \leftarrow 0$ 
15     break
16   else:
17     sample  $x_t$  uniformly from  $C_t$ 
18      $X \leftarrow$  used union  $\{x_1, x_2, \dots, x_t\}$ 
19      $W_b \leftarrow W_b * |C_t|$ 
20     return  $W_b$ 
21   5. Return labeled estimate  $\text{mean}(W_b)$  and unlabeled estimate
     $\hookrightarrow \text{mean}(W_b)/a_H$ .
22

```

Appendix D Stage 2: Subgraph Estimation in Null Model

This section provides the design of the second-stage estimation strategy not detailed in the main text, including the corresponding directed case, the proofs of the propositions, and the pipeline of the algorithm designed in the main text.

D.1 Design and Proof

The null model $\mathcal{G}_0$ for a given graph is given by a probability matrix $\mathcal{P} \in [0, 1]^{n \times n}$, which likewise satisfies: (1) symmetry, i.e., $p_{ij} = p_{ji}$; and (2) zero diagonal entries, i.e., $p_{ii} = 0$. Edges e_{ij} in the null model are generated independently. Two types of generations for random networks are usually available: (1) Erdős–Rényi type, for which $p_{ij} = \hat{p}_G$ for all $i < j$; and (2) Chung–Lu type, for which $p_{ij} = \min \left\{ \frac{d_i d_j}{2m}, 1 - \varepsilon \right\}$, where d_i is the undirected degree and m is the number of edges.

In the directed case, the Erdős–Rényi type sets $p_{ij} = \hat{p}_G$ for all $i \neq j$; the Chung–Lu type sets $p_{ij} = \min \{ d_i^{\text{out}} d_j^{\text{in}} / m, 1 - \varepsilon \}$.

For a given embedding ϕ , whether that embedding appears in a particular random network G^* is represented by a Bernoulli indicator variable $I_\phi(G^*)$. Under the population distribution $\mathcal{G}_0$ of the null model, conditional Monte Carlo strategy allows us to consider the occurrence probability

$p_H(\phi; \mathcal{G}_0) = \mathbb{E}[I_\phi(G^*)]$. The expected number of labeled occurrences of the target induced subgraph H , denoted by $M_H(\mathcal{G}_0)$, can then be computed as:

$$M_H(\mathcal{G}_0) = \mathbb{E}_{\mathcal{G}_0} [L_H(G^*)] = \mathbb{E}_{\mathcal{G}_0} \left[\sum_{\phi} I_\phi(G^*) \right] = \sum_{\phi} \mathbb{E}_{\mathcal{G}_0} [I_\phi(G^*)] = \sum_{\phi} \mathbb{P}_{\mathcal{G}_0} [\phi \text{ forms } H] \quad (\text{D.26})$$

where $\mathbb{P}_{\mathcal{G}_0} [\phi \text{ forms } H]$ is the probability that a given induced embedding ϕ forms the target induced subgraph under the null model, which can be expressed (1) in the undirected case as:

$$\mathbb{P}_{\mathcal{G}_0} [\phi \text{ forms } H] = \prod_{\{a,b\}: a < b} p_{\phi(a), \phi(b)}^{H_{a,b}} (1 - p_{\phi(a), \phi(b)})^{1-H_{a,b}} \quad (\text{D.27})$$

and (2) in the directed case as the product taken over all ordered pairs (a, b) with $a \neq b$:

$$\mathbb{P}_{\mathcal{G}_0} [\phi \text{ forms } H] = \prod_{(a,b): a \neq b} p_{\phi(a), \phi(b)}^{H_{a,b}} (1 - p_{\phi(a), \phi(b)})^{1-H_{a,b}} \quad (\text{D.28})$$

The unlabeled expectation is therefore $\mu_H(\mathcal{G}_0) = \frac{M_H(\mathcal{G}_0)}{a_H}$. Direct summation still requires traversal of $O(n^k)$ embeddings. Therefore, we again use the ISS estimator at this stage. It is worth

noting that the sampling weight no longer comes from the size of the candidate set; instead, it comes from the probability contribution of each candidate vertex relative to the sampled vertexes at each step. Let the ordering of subgraph vertexes be $\mathbf{o} = (o_1, o_2, \dots, o_k)$. At step t , the mapping of $(o_1, o_2, \dots, o_{t-1})$ has been completed, and the probability contribution associated with drawing candidate vertex x is denoted by $g_t(x)$. In the undirected case, $g_t(x)$ can be expressed as:

$$g_t(x) = \prod_{s < t} p_{x, x_s}^{H_{o_t, o_s}} (1 - p_{x, x_s})^{1 - H_{o_t, o_s}} \quad (\text{D.29})$$

and in the directed case as:

$$g_t(x) = \prod_{s < t} p_{x, x_s}^{H_{o_t, o_s}} (1 - p_{x, x_s})^{1 - H_{o_t, o_s}} p_{x_s, x}^{H_{o_s, o_t}} (1 - p_{x_s, x})^{1 - H_{o_s, o_t}} \quad (\text{D.30})$$

Let $Z_t = \sum_{x \notin \{x_1, x_2, \dots, x_{t-1}\}} g_t(x)$, and draw $x_t = x$ with probability $q_t(x) = \frac{g_t(x)}{Z_t}$, the log weight of a successful sampling path will be $\log W_0 = \sum_t \log Z_t$. Then,

Corollary 1 (Labeled count). The expectation of this weight is exactly $M_H(\mathbb{G}_0)$.

Proof. For the ordered tuple $\mathbf{x} = (x_1, \dots, x_k)$, the path weight is $W_0(\mathbf{x}) = \prod_{t=1}^k Z_t$. By using

$q_t(x) = \frac{g_t(x)}{Z_t}$, we have:

$$q(\mathbf{x})W_0(\mathbf{x}) = \left[\prod_{t=1}^k \frac{g_t(x_t)}{Z_t} \right] \left[\prod_{t=1}^k Z_t \right] = \prod_{t=1}^k g_t(x_t) \quad (\text{D.31})$$

therefore,

$$\mathbb{E}_q[W_0] = \sum_{\mathbf{x}} q(\mathbf{x})W_0(\mathbf{x}) = \sum_{\mathbf{x}} \prod_{t=1}^k g_t(x_t) \quad (\text{D.32})$$

According to the construction of $g_t(x)$, we have:

$$\prod_{t=1}^k \prod_{(o_s, o_t), \substack{\forall o_s \neq o_t \\ s < t}} p_{x, x_s}^{H_{o_t, o_s}} (1 - p_{x, x_s})^{1 - H_{o_t, o_s}} = \mathbb{P}_{\mathcal{G}_0}[(x_1, x_2, \dots, x_k) \text{ forms } H] \quad (\text{D.33})$$

and thus,

$$\mathbb{E}_q[W_0] = \sum_{\mathbf{x}} \mathbb{P}_{\mathcal{G}_0}[(x_1, x_2, \dots, x_k) \text{ forms } H] = M_H(\mathcal{G}_0) \quad (\text{D.34})$$

Q. E. D.

□

Hence, we can define the labeled and unlabeled subgraph-count estimators in the null model as:

$$\begin{cases} \widehat{M}_H(\mathcal{G}_0) = \frac{1}{R} \sum_{r=1}^R W_{0,r} \\ \widehat{N}_H(\mathcal{G}_0) = \frac{\widehat{M}_H(\mathcal{G}_0)}{a_H} \end{cases} \quad (\text{D.35})$$

where the properties of these two estimators are the same as those stated in the previous text.

D.2 Implementation

Here, we also give the pipeline of implementing the ISS algorithm for estimating subgraph within a null-model population distribution.

Algorithm D.2: ISS algorithm for null model.

```

1      Input:  $n$ -vertex null model with probability matrix  $\mathcal{P}$ ,  $k$ -
2      ↪ vertex subgraph  $H$ , sample times  $R$ , and rare-constraint order  $o$ 
3      ↪  $= (o_1, o_2, \dots, o_k)$ .
      Output: count of subgraph  $H$ .
```

```

4      1. Normalize  $A$  and  $H$  as simple undirected zero-diagonal
      ↪ matrices.
5      2. Build bitsets and conduct sampling via:
6      for  $r = 1, 2, \dots, R$ :
7       $X \leftarrow \text{empty}$ 
8      log  $W_r \leftarrow 0$ 
9      for  $t = 1, 2, \dots, k$ :
10     for each unused candidate  $x$ , compute incremental probability
      ↪  $g_t(x)$ 
11      $Z_t \leftarrow \text{sum}(g_t(x))$ 
12     if  $Z_t$  is empty:
13     break
14     else:
15     sample  $x_t$  with probability  $g_t(x)/Z_t$ 
16      $X \leftarrow \text{used union } \{x_1, x_2, \dots, x_t\}$ 
17     log( $W_r$ )  $\leftarrow$  log( $W_r$ ) + log( $Z_t$ )
18     return log( $W_r$ )
19     3. Return labeled null expectation mean(exp(log( $W_r$ ))).
20     4. Divide by automorphism count  $a_H$  to obtain the unlabeled
      ↪ null expectation.

```

Appendix E Motif Verification

This section provides the strategy for motif verification which is not detailed in the main text, including the corresponding directed case and the proofs for propositions.

E.1 Motif Verification Design and Proof

Based on the two-stage estimation described above, we obtain both estimates of the subgraph count and the expected one in the given and random networks. We now consider statistically testing via statistics construction, in order to examine whether the number of target subgraphs in the targeted network is significantly different from (larger than) that of the null model. This determines whether the target subgraph can be regarded as a genuine motif in the targeted network.

Denote $\theta = N_H(G)$ as the subgraph count in the real network, $\mu_0 = \mathbb{E}[N_H(G^*)]$ as the null-model expectation, and $\tau_0^2 = \mathbb{V}[N_H(G^*)]$ as the null-model variance. Here we construct the log ratio $\beta = \log\left(\frac{\theta}{\mu_0}\right)$, and refer to it as the subgraph enrichment coefficient β_{enr} . The corresponding estimator is $\hat{\beta} = \log\left(\frac{\hat{\theta}_B}{\hat{\mu}_{0,R}}\right)$, which can be computed from $\hat{\theta}_B$ and $\hat{\mu}_{0,R}$, obtained respectively from the two stages of estimation. Then, we have the following propositions:

Proposition 3 (Consistency). $\widehat{\beta}$ is a consistent estimator of β , and $s_{\widehat{\beta}}^2 = \frac{s_{\widehat{\theta}}^2}{B\widehat{\theta}_B^2} + \frac{s_{\widehat{\mu}}^2}{R\widehat{\mu}_R^2}$ is also a consistent estimator of $\mathbb{V}[\widehat{\beta}]$.

Proof. First, we will show that, $\widehat{\beta}$ is a consistent estimator of β . From the preceding deduction, since $B \rightarrow \infty$, the SLLN gives:

$$\widehat{\theta}_B \xrightarrow{a.s.} \theta \quad (\text{E.36})$$

Similarly, since $R \rightarrow \infty$, the SLLN also gives:

$$\frac{1}{R} \sum_{r=1}^R W_{0,r} \xrightarrow{a.s.} \mathbb{E}[W_{0,r}] \quad (\text{E.37})$$

According to the construction of $\widehat{\mu}_{0,R}$, we have:

$$\widehat{\mu}_{0,R} \xrightarrow{a.s.} \mathbb{E} \left[\frac{W_{0,r}}{a_H} \right] = \mathbb{E}[N_H(G^*)] = \mu_0 \quad (\text{E.38})$$

Since $\widehat{\mu}_{0,R} > 0$, the Continuous Mapping Theorem implies:

$$\widehat{\beta} = \log \left(\frac{\widehat{\theta}_B}{\widehat{\mu}_R} \right) \xrightarrow{a.s.} \log \left(\frac{\theta}{\mu} \right) = \beta \quad (\text{E.39})$$

therefore, $\widehat{\beta}$ is a consistent estimator of β .

Worthy mentioning, $\widehat{\beta}$ is not a strictly unbiased estimator of β , however, it is still asymptotic unbiased. For $\log(\cdot)$ being strictly concave, Jensen's inequality gives:

$$\mathbb{E}[\log \widehat{\theta}_B] \leq \log \mathbb{E}[\widehat{\theta}_B] = \log \theta_B \quad (\text{E.40})$$

and similarly, we have: $\mathbb{E}[\log \widehat{\mu}_{0,R}] \leq \log \mathbb{E}[\widehat{\mu}_{0,R}] = \log \mu_{0,R}$. Thus, $\text{Bias}[\widehat{\beta}] = \mathbb{E}[\widehat{\beta}] - \beta = (\mathbb{E}[\log \widehat{\theta}_B] - \log \theta_B) - (\mathbb{E}[\log \widehat{\mu}_{0,R}] - \log \mu_{0,R})$, which is generally not equal to 0. Hence, $\widehat{\beta}$ is not a strictly unbiased estimator of β .

Taking a second-order Taylor expansion of $\mathbb{E}[\log \widehat{\theta}]$, we have:

$$\mathbb{E}[\log \widehat{\theta}] \approx \log \theta - \frac{\mathbb{V}[\widehat{\theta}]}{2\theta^2} \quad (\text{E.41})$$

and similarly,

$$\mathbb{E}[\log \widehat{\mu}] \approx \log \mu - \frac{\mathbb{V}[\widehat{\mu}]}{2\mu^2} \quad (\text{E.42})$$

therefore,

$$\mathbb{E}[\widehat{\beta}] \approx \log(\theta/\mu) - \frac{\mathbb{V}[\widehat{\theta}]}{2\theta^2} + \frac{\mathbb{V}[\widehat{\mu}]}{2\mu^2} \quad (\text{E.43})$$

Since $\begin{cases} \mathbb{V}[\widehat{\theta}] = \sigma_\theta^2/B \\ \mathbb{V}[\widehat{\mu}] = \sigma_\mu^2/R \end{cases}$, we have: $\text{Bias}[\widehat{\beta}] \approx -\frac{\sigma_\theta^2}{2B\theta^2} + \frac{\sigma_\mu^2}{2R\mu^2}$. Thus, as $B, R \rightarrow \infty$,

$$\mathbb{E}[\widehat{\beta}] \rightarrow \log\left(\frac{\theta}{\mu}\right) = \beta. \quad (\text{E.44})$$

Second, we will show that, $s_\beta^2 = \frac{s_\theta^2}{B\theta_B^2} + \frac{s_\mu^2}{R\widehat{\mu}_R^2}$ is a consistent estimator of $\mathbb{V}[\widehat{\beta}]$. According to definition, $s_\theta^2 = \frac{1}{B-1} \sum_b (W_b - \widehat{\theta}_B)^2$ and $s_\mu^2 = \frac{1}{R-1} \sum_r (W_{0,r} - \widehat{\mu}_{0,R})^2$. The preceding deduction has shown that, $s_\theta^2 \xrightarrow{a.s.} \sigma_\theta^2$. Similarly, the SLLN gives:

$$\begin{cases} \frac{1}{R} \sum_{r=1}^R W_{0,r} \rightarrow \mathbb{E}[W_{0,r}] \\ \frac{1}{R} \sum_{r=1}^R W_{0,r}^2 \rightarrow \mathbb{E}[W_{0,r}^2] \end{cases} \quad (\text{E.45})$$

therefore,

$$\left(\frac{1}{R} \sum_{r=1}^R W_{0,r} \right)^2 \xrightarrow{a.s.} (\mathbb{E}[W_{0,r}])^2 \quad (\text{E.46})$$

Since $\lim_{R \rightarrow \infty} \frac{R}{R-1} = 1$, we have:

$$s_\mu^2 \xrightarrow{a.s.} \mathbb{E}[W_{0,r}^2] - (\mathbb{E}[W_{0,r}])^2 = \mathbb{V}[W_{0,r}] = \sigma_\mu^2 \quad (\text{E.47})$$

thus, s_μ^2 is a consistent estimator of σ_μ^2 . According to the Continuous Mapping Theorem, we then have:

$$s_\beta^2 = \frac{s_\theta^2}{B\hat{\theta}_B^2} + \frac{s_\mu^2}{R\hat{\mu}_R^2} \rightarrow \frac{\sigma_\theta^2}{B\theta_B^2} + \frac{\sigma_\mu^2}{R\mu_R^2} = \mathbb{V}[\hat{\beta}] \quad (\text{E.48})$$

Q. E. D. □

Proposition 4 (Asymptotic normality). $\hat{\beta} - \beta$ is asymptotically normally distributed.

Proof. In a fixed finite network, $\mathbb{E}[W] < \infty$, $\mathbb{V}[W] < \infty$, $\mathbb{E}[W_0] < \infty$, $\mathbb{V}[W_0] < \infty$. Therefore, the CLT gives:

$$\begin{cases} \sqrt{B}(\hat{\theta}_B - \theta) \Rightarrow N(0, \sigma_\theta^2) \\ \sqrt{R}(\hat{\mu}_R - \mu_0) \Rightarrow N(0, \sigma_\mu^2) \end{cases} \quad (\text{E.49})$$

Since the first-stage and second-stage sampling procedures are independent, $\begin{pmatrix} \hat{\theta}_B - \theta \\ \hat{\mu}_R - \mu_0 \end{pmatrix} \overset{\text{approx}}{\sim} N\left(0, \begin{pmatrix} \sigma_\theta^2/B & 0 \\ 0 & \sigma_\mu^2/R \end{pmatrix}\right)$. The gradient of $\beta(\theta, \mu)$ is $\nabla\beta = \begin{pmatrix} \partial\beta/\partial\theta \\ \partial\beta/\partial\mu \end{pmatrix} = \begin{pmatrix} 1/\theta \\ -1/\mu \end{pmatrix}$.

By the delta method,

$$\widehat{\beta} - \beta \approx \frac{1}{\theta}(\widehat{\theta}_B - \theta) - \frac{1}{\mu}(\widehat{\mu}_R - \mu_0) \quad (\text{E.50})$$

hence,

$$\mathbb{V}[\widehat{\beta}] \approx \frac{1}{\theta^2} \frac{\sigma_\theta^2}{B} + \frac{1}{\mu^2} \frac{\sigma_\mu^2}{R} \quad (\text{E.51})$$

According to the CLT, we have:

$$\widehat{\beta} - \beta \stackrel{\text{approx}}{\sim} N\left(0, \frac{1}{\theta^2} \frac{\sigma_\theta^2}{B} + \frac{1}{\mu^2} \frac{\sigma_\mu^2}{R}\right) \quad (\text{E.52})$$

Q. E. D. □

For a population null-model distribution $\mathcal{P}_0$, any null random network G^* follows the null-model distribution, i.e., $G^* \sim \mathcal{P}_0$. The true number of targeted subgraph in the random network is $N_H(G^*)$, following the distribution $\mathcal{L}\{N_H(G^*) : G^* \sim \mathcal{P}_0\}$. The true number of targeted subgraph in the random network has expectation $\mu_0 = \mathbb{E}[N_H(G^*)]$ and variance $\tau_0^2 = \mathbb{V}[N_H(G^*)]$. Meanwhile, we assume that the subgraph count under the null model can be approximated by a Poisson distribution:

Assumption 1 (Poisson approximation). $N_H(G^*) \stackrel{\text{approx}}{\sim} \text{Poisson}(\mu_0)$, so that $\tau_0^2/\mu_0 \rightarrow 1$.

To construct a simple z -test, normality is required, which, however, requires $\mu_0 \rightarrow \infty$. Here, we therefore impose the following finite-sample approximation:

Assumption 2 (Poisson-to-normal approximation). In finite samples, μ_0 is sufficiently large to be approximated by a normal distribution, i.e., $N_H(G^*) \stackrel{\text{approx}}{\sim} N(\mu_0, \mu_0)$.²

Now, we can construct a z -statistic and test whether $\hat{\beta}$ is significantly greater than zero, thereby determining whether a particular subgraph can be regarded as a genuine motif. Specifically, we have the following proposition:

Proposition 5 (z -statistic). For the null hypothesis $H_0 : G \sim \mathcal{P}_0$, equivalently $N_H(G) = \mathbb{E}[N_H(G^*)]$, and the right-tail alternative $H_1 : N_H(G) > \mathbb{E}[N_H(G^*)]$, construct the z -statistic $Z_\beta = \left[s_\theta^2 (B\hat{\theta}^2)^{-1} + s_\mu^2 (R\hat{\mu}_0^2)^{-1} + \hat{\mu}_0^{-1} \right]^{-1/2} \log \left(\frac{\hat{\theta}}{\hat{\mu}_0} \right)$. Then $Z_\beta \stackrel{\text{approx}}{\sim} N(0, 1)$.

Proof. Construct the generalized z -statistic by:

$$Z_\beta = \frac{\log(\hat{\theta}/\hat{\mu}_0) - \log(\theta/\mu_0)}{\sqrt{\mathbb{V}[\log \hat{\theta} - \log \hat{\mu}_0]}} \quad (\text{E.53})$$

²In practice, the normal approximation can be used since $\mu_0 \geq 15$.

Let $\begin{cases} \hat{\theta} = \theta + \varepsilon_\theta \\ \hat{\mu}_0 = \mu_0 + \varepsilon_\mu \end{cases}$, then $\hat{\beta}$ can be rewritten as:

$$\hat{\beta} = \log \left(\frac{\theta + \varepsilon_\theta}{\mu_0 + \varepsilon_\mu} \right) = \log \left(1 + \frac{\theta - \mu_0}{\mu_0} \right) + \log \left(1 + \frac{\varepsilon_\theta}{\theta} \right) - \log \left(1 + \frac{\varepsilon_\mu}{\mu_0} \right) \quad (\text{E.54})$$

Under the null hypothesis and the Poisson approximation, the three terms above can be expanded to their first orders:

$$\begin{cases} \log \left(1 + \frac{\theta - \mu_0}{\mu_0} \right) = \frac{\theta - \mu_0}{\mu_0} + o_p(\mu_0^{-1/2}) \\ \log \left(1 + \frac{\varepsilon_\theta}{\theta} \right) = \frac{\varepsilon_\theta}{\theta} + o_p(\mathbb{V}_\theta^{1/2}/\theta) \\ \log \left(1 + \frac{\varepsilon_\mu}{\mu_0} \right) = \frac{\varepsilon_\mu}{\mu_0} + o_p(\mathbb{V}_\mu^{1/2}/\mu_0) \end{cases} \quad (\text{E.55})$$

therefore,

$$\hat{\beta} = \frac{\theta - \mu_0}{\mu_0} + \frac{\varepsilon_\theta}{\theta} - \frac{\varepsilon_\mu}{\mu_0} + o_p \left(\sqrt{\frac{1}{\mu_0} + \frac{\mathbb{V}_\theta}{\theta^2} + \frac{\mathbb{V}_\mu}{\mu_0^2}} \right) \quad (\text{E.56})$$

Since the sampling procedures are independent, $\text{Cov}_0(T - \mu_0, \varepsilon_\mu) = 0$ and $\text{Cov}_0(\varepsilon_\theta, \varepsilon_\mu) = 0$. Moreover, since $\mathbb{E}[\varepsilon_\theta \mid G] = \mathbb{E}[\hat{\theta} - \theta \mid G] = \mathbb{E}[\hat{\theta} \mid G] - \mathbb{E}[\theta \mid G] = 0$, we have $\text{Cov}(T, \varepsilon_\theta) =$

$\mathbb{E}[T\varepsilon_\theta] - \mathbb{E}[T]\mathbb{E}[\varepsilon_\theta] = \mathbb{E}[T\mathbb{E}[\varepsilon_\theta | G]] + \mathbb{E}[\mathbb{E}[\varepsilon_\theta | G]] = 0$. Thus, the variance of $\hat{\beta}$ can be written as:

$$\mathbb{V}[\hat{\beta}] \approx \mathbb{V}\left[\frac{\theta - \mu_0}{\mu_0}\right] + \mathbb{V}\left[\frac{\varepsilon_\theta}{\theta}\right] + \mathbb{V}\left[\frac{\varepsilon_\mu}{\mu_0}\right] = \frac{1}{\mu_0} + \frac{\mathbb{V}_\theta}{\theta^2} + \frac{\mathbb{V}_\mu}{\mu_0^2} \quad (\text{E.57})$$

Under the null hypothesis, the z -statistic is therefore:

$$Z_\beta = \frac{\hat{\beta}}{\sqrt{\mathbb{V}[\hat{\beta}]}} = \frac{\log(\hat{\theta}/\hat{\mu}_0)}{\sqrt{\frac{\mathbb{V}_\theta}{\theta^2} + \frac{\mathbb{V}_\mu}{\mu_0^2} + \frac{1}{\mu_0}}} \quad (\text{E.58})$$

By the CLT, Z_β is asymptotically distributed as $N(0, 1)$. Since the population variances are unknown, employ $\hat{\mathbb{V}}_\theta = \frac{s_\theta^2}{B}$, $\hat{\mathbb{V}}_\mu = \frac{s_\mu^2}{R}$, together with $\hat{\theta}$ and $\hat{\mu}_0$ as the plug-in estimators. Based on the almost sure convergence conditions: $s_\theta^2 \xrightarrow{a.s.} \sigma_\theta^2$, $s_\mu^2 \xrightarrow{a.s.} \sigma_\mu^2$, $\hat{\theta} \xrightarrow{a.s.} \theta$, and $\hat{\mu}_0 \xrightarrow{a.s.} \mu_0$, we have at least:

$$\frac{s_\theta^2}{B\hat{\theta}^2} + \frac{s_\mu^2}{R\hat{\mu}_0^2} + \hat{\mu}_0^{-1} \xrightarrow{p} \mathbb{V}[\hat{\beta}] \quad (\text{E.59})$$

then, Slutsky's theorem gives:

$$Z_\beta = \frac{\log(\hat{\theta}/\hat{\mu}_0)}{\sqrt{\frac{s_\theta^2}{B\hat{\theta}^2} + \frac{s_\mu^2}{R\hat{\mu}_0^2} + \hat{\mu}_0^{-1}}} \xrightarrow{d} N(0, 1) \quad (\text{E.60})$$

Q. E. D.

□

E.2 Pólya–Aeppli distribution and Poisson degeneration

The distribution of a specific subgraph in a random graph can be described by an approximated Pólya–Aeppli (PA) distribution (also called geometric Poisson distribution). Under certain conditions, the PA distribution can degenerate into a non-degenerate Poisson distribution (i.e., $\text{Poisson}(\lambda)$ with $\lambda > 0$). Here, we investigate the conditions that a random graph should satisfy for the PA distribution to degenerate into a Poisson distribution. Actually, we have the following proposition:

Proposition 6 (Poisson degeneration). Within the Erdős–Rényi null model, once p_n is of the same order as $n^{-2/(k-1)}$, namely, $p_n \asymp n^{-2/(k-1)}$, the PA distribution degenerates into a Poisson distribution, namely, $\text{PA}(\Lambda_n, a_n) \xrightarrow{d} \text{Poisson}(\lambda)$.

Proof. Let a random graph follow an Erdős–Rényi-type random graph distribution, namely, $G_n \sim \text{ER}(n, p_n)$; and let X_k denote the count of k -cliques within G_n . The PA approximation uses matching of the first two moments, namely, $X_{k,n} \sim \text{PA}(\Lambda_n, a_n)$. Thereinto, $a_n = \frac{\sigma_n^2 - \mu_n}{\sigma_n^2 + \mu_n}$ and $\Lambda_n = (1 - a_n) \mu_n$, where $\mu_n = \mathbb{E}[X_{k,n}]$ and $\sigma_n^2 = \mathbb{V}[X_{k,n}]$. For the PA distribution degenerating into the non-degenerate Poisson distribution, i.e., $\text{PA}(\Lambda_n, a_n) \xrightarrow{d} \text{Poisson}(\lambda)$, $0 < \lambda < \infty$, it

requires two conditions: $a_n \rightarrow 0$ and $\Lambda_n \rightarrow \lambda$. Since $a_n \rightarrow 0$ ensures $\Lambda_n = (1 - a_n) \mu_n \sim \mu_n$, it is therefore necessary that $\mu_n \rightarrow \lambda$.

First consider the scale condition for the degeneration of Poisson distribution. Such a condition can be derived by examining the count expectation of k -cliques. For any k -vertex subset $S \subseteq V_G$, define an indicator function for the realization of a k -clique in G_n , namely, $I_S = \mathbb{1}\{G_n[S] \text{ forms a } k\text{-clique}\}$. Then the count of k -cliques in G_n should be $X_k = \sum_{S \subseteq V_G, |S|=k} I_S$, whose expectation can be expressed as:

$$\mathbb{E}[X_k] = \sum_{S \subseteq V_G, |S|=k} \mathbb{E}[I_S] = \sum_{S \subseteq V_G, |S|=k} \mathbb{P}[G_n[S] \text{ forms a } k\text{-clique}] \quad (\text{E.6r})$$

Denote the number of edges within a k -clique as $e_k = \binom{k}{2} = \frac{k(k-1)}{2}$. Since edges in an Erdős–Rényi-type random graph are independent, the probability that the subgraph $G_n[S]$ induced by the vertex subset S forms a k -clique will exactly be the probability that all edges of the k -clique are jointly and independently realized, that is, $\mathbb{P}[G_n[S] \text{ forms a } k\text{-clique}] = p_n^{e_k}$. Therefore, the expectation of the k -clique count will be $\mu_n = \mathbb{E}[X_k] = \binom{n}{k} p_n^{e_k}$. Accordingly, the non-degenerate Poisson distribution requires $\binom{n}{k} p_n^{e_k} \rightarrow \lambda$. By $\binom{n}{k} = \frac{n^k}{k!}$, we have $n^{k/e_k} p_n \rightarrow (k!\lambda)^{1/e_k}$; and by $e_k = \frac{k(k-1)}{2}$, we further have $\frac{k}{e_k} = \frac{2}{k-1}$, and hence $p_n n^{2/(k-1)} \rightarrow (k!\lambda)^{1/\binom{k}{2}}$.

Thus, for fixed k and λ , let $c = (k!\lambda)^{1/\binom{k}{2}}$, the condition for p_n can be written as $p_n \sim$

$cn^{-2/(k-1)}$; or, it can be denoted as $p_n \asymp n^{-2/(k-1)}$. Here, this condition can be also called the “scale condition”.

Second, consider the condition $a_n \rightarrow 0$. In fact, such a condition can be automatically implied by the scale condition mentioned above. Here, designate the dispersion parameter D_n as $D_n = \frac{\sigma_n^2}{\mu_n}$, then $a_n = \frac{D_n - 1}{D_n + 1}$. Therefore, we have $a_n \rightarrow 0 \iff D_n \rightarrow 1$.

For a k -clique, its count variance can be expressed as:

$$\sigma_n^2 = \mu_n (1 - p_n^{e_k}) + \sum_{s=2}^{k-1} \binom{n}{k} \binom{k}{s} \binom{n-k}{k-s} \left(p_n^{2e_k - \binom{s}{2}} - p_n^{2e_k} \right) \quad (\text{E.62})$$

thus,

$$D_n = 1 - p_n^{e_k} + \sum_{s=2}^{k-1} \binom{k}{s} \binom{n-k}{k-s} p_n^{e_k - \binom{s}{2}} \left(1 - p_n^{\binom{s}{2}} \right) \quad (\text{E.63})$$

For the PA distribution degenerating into a Poisson distribution, the corresponding overlapping terms must disappear, that is, for $s = 2, \dots, k-1$, $\binom{n-k}{k-s} p_n^{e_k - \binom{s}{2}} \rightarrow 0$, or equivalently, $n^{k-s} p_n^{\binom{k}{2} - \binom{s}{2}} \rightarrow 0$. By $p_n \asymp n^{-2/(k-1)}$, we actually have $n^{k-s} p_n^{\binom{k}{2} - \binom{s}{2}} \asymp n^{\frac{s(s-k)}{k-1}}$. For $2 \leq s \leq$

$k - 1$, we also have $\frac{s(s-k)}{k-1} < 0$. Hence, since $n \rightarrow \infty$, the overlapping contribution tends to be 0, i.e., $n^{\frac{s(s-k)}{k-1}} \rightarrow 0$; accordingly, $D_n \rightarrow 1$, $a_n \rightarrow 0$.

Q. E. D. □

Now, we consider the case with the Chung-Lu-type random graph as the null model. Actually, we also have the following proposition:

Proposition 7 (Poisson degeneration). Within the Chung-Lu null model, since $A_{k-1,n}$ is of the same order as $W_n^{-\frac{2}{(k-1)}}$ and $\frac{A_{2k-s-1,n}^s A_{k-1,n}^{2(k-s)}}{W_n^{2e_k - \binom{s}{2}}} \rightarrow 0$, the PA distribution can be seen to degenerate into a Poisson distribution.

Proof. Let a random graph follow a Chung-Lu-type distribution, namely, $G_n \sim \text{CL}(n, W)$, where

$W = \begin{pmatrix} w_{11} & w_{12} & \cdots & w_{1n} \\ w_{21} & w_{22} & \cdots & w_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ w_{n1} & w_{n2} & \cdots & w_{nn} \end{pmatrix}$ is the weight matrix. Edge realization will be mutually independent,

and the probability that realizing any edge e_{ij} is $\mathbb{P}(A_{ij} = 1) = p_{ij} = \frac{w_{ni}w_{nj}}{W_n}$, where $W_n = \sum_{i=1}^n w_{ni}$. In addition, to avoid cap effects, we let $\max_{i < j} \frac{w_{ni}w_{nj}}{W_n} \rightarrow 0$. This condition can also be

referred to as the “non-domination condition”, for the realization probability of any edge depending only on vertex weights and converging to 0, rather than being realized with probability 1. And under such a condition, after imposing the probability upper-bound truncation $p_{ij} = \min \left\{ \frac{w_i w_j}{W}, 1 \right\}$, the subgraph expectation and variance both retain the full information of degree moments.

Similarly, let X_k denote the number of k -cliques in G_n , and use the matching of the first two moments as the Pólya–Aeppli approximation, namely, $X_{k,n} \sim \text{PA}(\Lambda_n, a_n)$; here, the PA distribution is obtained by matching the mean and variance, namely, $a_n = \frac{\sigma_n^2 - \mu_n}{\sigma_n^2 + \mu_n}$ and $\Lambda_n = (1 - a_n) \mu_n$; where $\mu_n = \mathbb{E}[X_{k,n}]$ and $\sigma_n^2 = \text{Var}(X_{k,n})$. As stated above, for PA degenerating into a non-degenerate Poisson distribution, i.e., $\text{PA}(\Lambda_n, a_n) \xrightarrow{d} \text{Poisson}(\lambda)$, $0 < \lambda < \infty$, it likewise requires $a_n \rightarrow 0$, $\Lambda_n \rightarrow \lambda$; and since when $a_n \rightarrow 0$ we have $\Lambda_n \sim \mu_n$, it is therefore first necessary that $\mu_n \rightarrow \lambda$.

First consider the scale condition for degeneration. This condition can likewise be derived by investigating the count expectation of k -cliques. For any k -vertex set $S \subseteq V_G$, define the indicator function $I_S = \mathbb{1} \{G_n[S] \text{ forms a } k\text{-clique}\}$, and we still have the k -clique count $X_{k,n} = \sum_{|S|=k} I_S$. However, under the Chung–Lu null model, the expectation of the k -clique count will be:

$$\mathbb{E}[I_S] = \sum_{|S|=k} \mathbb{P}\{G_n[S] \text{ forms a } k\text{-clique}\} = \prod_{\{i,j\} \subset S} \frac{w_{ni} w_{nj}}{W_n} \quad (\text{E.64})$$

and therefore,

$$\mu_n = W_n^{-e_k} \sum_{|S|=k} \prod_{i \in S} w_{ni}^{k-1} \quad (\text{E.65})$$

Accordingly, the first condition for the degeneration of Poisson distribution will be $W_n^{-e_k} \sum_{|S|=k} \prod_{i \in S} w_{ni}^{k-1} \rightarrow \lambda$. Under the preceding assumption $\max_{i < j} \frac{w_{ni} w_{nj}}{W_n} \rightarrow 0$, we have $\frac{\max_i w_{ni}^{k-1}}{\sum_i w_{ni}^{k-1}} \rightarrow 0$. Here, designate $A_{r,n} = \sum_{i=1}^n w_{ni}^r$, then $\sum_{|S|=k} \prod_{i \in S} w_{ni}^{k-1} \sim \frac{A_{k-1,n}^k}{k!}$. Thus the preceding condition can be rewritten as: $\frac{A_{k-1,n}^k}{k! W_n^{e_k}} \rightarrow \lambda$; or equivalently, $\frac{A_{k-1,n}}{W_n^{(k-1)/2}} \rightarrow (k! \lambda)^{1/k}$. Similarly, here, we can also designate $c = (k! \lambda)^{1/\binom{k}{2}}$, and then we will have $A_{k-1,n} \asymp W_n^{(k-1)/2}$; such a condition, in fact, corresponds to the scale condition in the Erdős–Rényi case.

Second, consider the overlapping condition for degeneration. Unlike the Erdős–Rényi model, this condition cannot be directly implied by the aforementioned scale condition. As stated above, for the PA distribution degenerating into a non-degenerate Poisson distribution, it also requires $a_n \rightarrow 0$. Since $a_n = \frac{D_n - 1}{D_n + 1}$, $D_n = \frac{\sigma_n^2}{\mu_n}$, the condition $a_n \rightarrow 0$ is, therefore, equivalent to $D_n \rightarrow 1$, i.e., $\sigma_n^2 \sim \mu_n$. Thus, the count variance of the k -clique here can be re-expressed as:

$$\sigma_n^2 = \sum_{S \subseteq V_G} \mathbb{V}(I_S) + \sum_{S, T \subseteq V_G, S \neq T} \text{Cov}(I_S, I_T) \quad (\text{E.66})$$

Since two k -cliques share fewer than two vertices, they share no edge and therefore are independent, in which case, their corresponding count covariance will be 0. This means that, only clique pairs sharing $s = 2, \dots, k - 1$ vertices will contribute overlapping count covariance. If $|S \cap T| = s$, then the joint realization probability $q_{S,T} = \mathbb{E}[I_S I_T]$ of I_S and I_T can be expressed as:

$$q_{S,T} = W_n^{-\{2e_k - \binom{s}{2}\}} \prod_{i \in S \cap T} w_{ni}^{2k-s-1} \prod_{j \in S \Delta T} w_{nj}^{k-1} \quad (\text{E.67})$$

By $0 \leq \mathbb{E}[I_S] \mathbb{E}[I_T] \leq q_{S,T}$, we have $0 \leq \text{Cov}(I_S, I_T) \leq q_{S,T}$, and hence $\sum_{S \neq T} \text{Cov}(I_S, I_T) \leq \sum_{s=2}^{k-1} \sum_{S,T} q_{S,T}$. Now, designate:

$$\Omega_{k,s,n} := W_n^{-\{2e_k - \binom{s}{2}\}} \sum_{\substack{|S|=|T|=k \\ |S \cap T|=s}} \prod_{i \in S \cap T} w_{ni}^{2k-s-1} \prod_{j \in S \Delta T} w_{nj}^{k-1} \quad (\text{E.68})$$

at this point, a sufficient overlapping condition for asymptotically following a non-degenerate Poisson distribution is that, for $s = 2, \dots, k - 1$, $\Omega_{k,s,n} \rightarrow 0$, which is also called the overlapping condition. In fact, since this condition holds, we have $\Omega_{k,s,n} := \sum_{S,T} q_{S,T} \rightarrow 0$, and therefore $\sum_{S \neq T} \text{Cov}(I_S, I_T) \leq \sum_{s=2}^{k-1} \Omega_{k,s,n} = o(1)$ and $\sigma_n^2 = \mu_n + o(1)$; hence, $D_n \rightarrow 1, a_n \rightarrow 0$.

Under the non-domination condition $\max_{i < j} \frac{w_{ni}w_{nj}}{W_n} \rightarrow 0$, the above overlapping condition can be rewritten in the moment-condition form, namely, for $s = 2, \dots, k-1$, $\frac{A_{2k-s-1,n}^s A_{k-1,n}^{2(k-s)}}{W_n^{2e_k - \binom{s}{2}}} \rightarrow 0$.

Q. E. D. □

Of course, here, we also report the performance of p -values computed using the Poisson approximation, namely their discrepancy from the empirical p -values. The p -value calculation results of triangle, 4-clique, and 5-clique motifs under different graph densities are shown in Figs.E.1, E.2 and E.3.³ The results indicate that, while graph density is relatively low, the p -values based on the Poisson approximation do exhibit relatively good performance.

³It is worth noting that, even in the case with lowest graph density, the simulated networks are constructed so that the count expectation of target subgraphs under the Erdős–Rényi-type random network is no less than 20, i.e., $\mathbb{E}[\hat{N}_{\mathcal{G}_0}] \geq 20$.

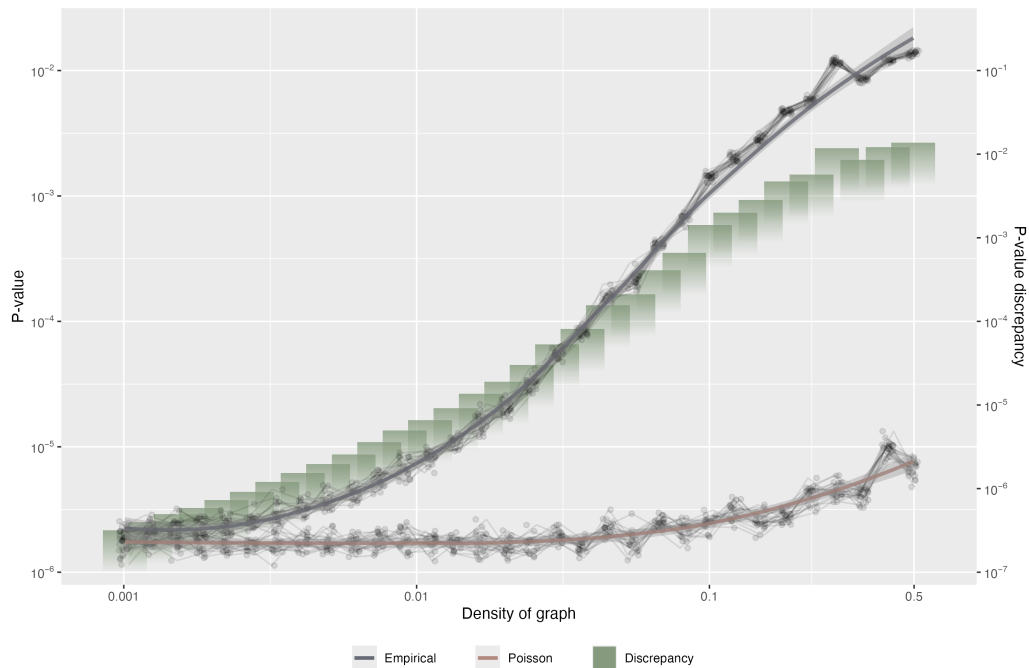


Figure E.1: Discrepancies between empirical p -values and Poisson approximation (triangle motif).

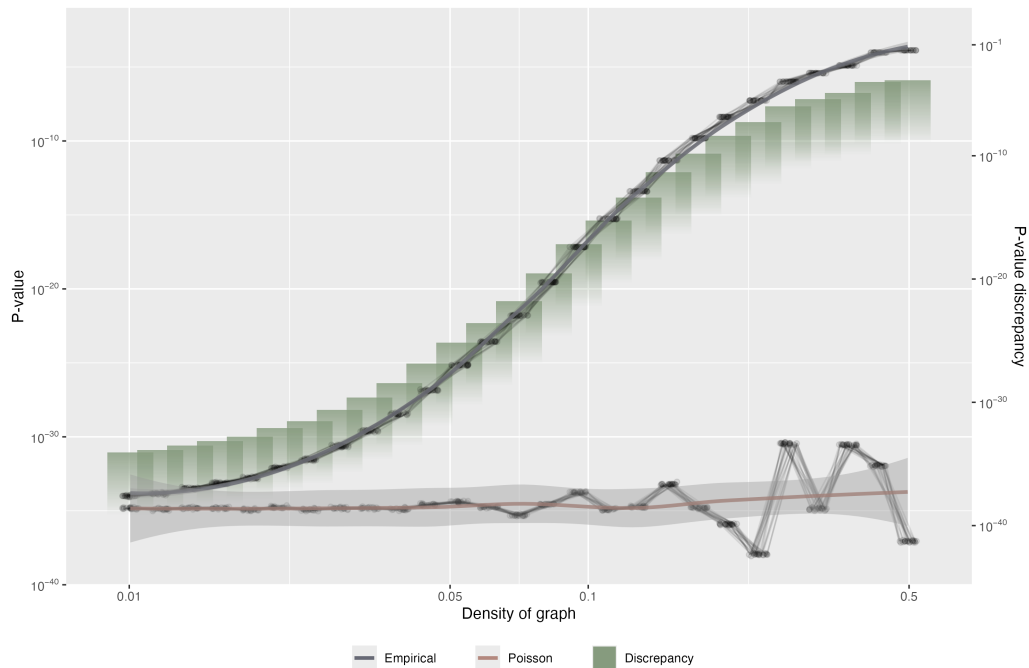


Figure E.2: Discrepancies between empirical p -values and Poisson approximation (4-clique motif).

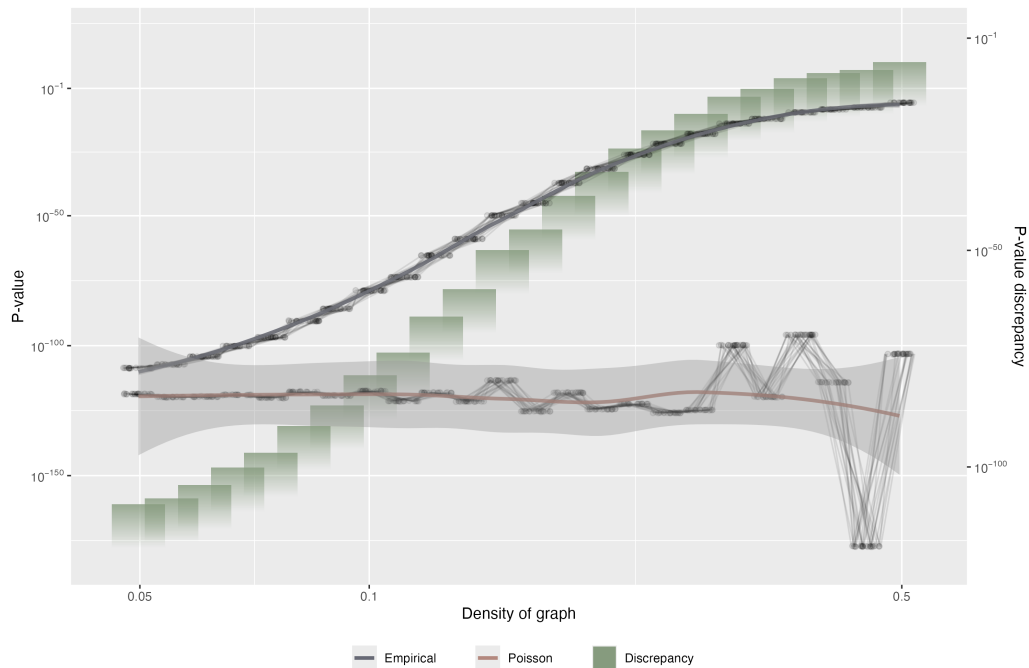


Figure E.3: Discrepancies between empirical p -values and Poisson approximation (5-clique motif).

E.3 Random-Network Calibration

The conditional Monte Carlo strategy based on direct null-expectation estimation does avoid computationally intensive bootstrap sampling. However, the preceding statistical test relies on the Poisson approximation and on the tail behavior under the induced constraints. Therefore, we further consider random network calibration based on bootstrap sampling. The procedure is as follows: First, generate K i.i.d. random networks from the population null-model distribution $\mathcal{G}_0$, i.e., $G_1^*, \dots, G_K^* \stackrel{i.i.d.}{\sim} \mathcal{G}_0$. Second, for each random network, estimate the target subgraph count using the ISS algorithm, i.e., $\hat{\theta}_k^* = \hat{N}_H(G_k^*)$. Then, using the same direct null-expectation estimator $\hat{\mu}_0$, construct the subgraph enrichment coefficient estimator for each random network, i.e., $\hat{\beta}_k^* = \log\left(\frac{\hat{\theta}_k^*}{\hat{\mu}_0}\right)$, and take the mean $\bar{\beta}^* = K^{-1} \sum_{k=1}^K \hat{\beta}_k^*$. Meanwhile, define the sample variance $\hat{\omega}_\beta^2 = K^{-1} \sum_{k=1}^K \left(\hat{\beta}_k^* - \bar{\beta}^*\right)^2$. We have the following propositions:

Proposition 8 (Consistency). $\bar{\beta}^*$ is a consistent estimator of $\mathbb{E}[\beta^*] = \mathbb{E}\left[\log\left(\frac{N_H(G^*)}{\mu_0}\right)\right]$, and $\hat{\omega}_\beta^2$ is also a consistent estimator of $\mathbb{V}[\beta^*] = \mathbb{V}\left[\log\left(\frac{N_H(G^*)}{\mu_0}\right)\right]$.

Proof. According to the preceding deduction, we obtain two almost sure convergence conditions:

$\hat{\theta}_k^* \xrightarrow{a.s.} \theta_k^*, \hat{\mu}_0 \xrightarrow{a.s.} \mu_0$. Therefore, as $K \rightarrow \infty$,

$$\bar{\beta}^* = \frac{1}{K} \sum_{k=1}^K \hat{\beta}_k^* \xrightarrow{a.s.} \frac{1}{K} \sum_{k=1}^K \beta_k^* \xrightarrow{a.s.} \beta^* = \log \left(\frac{N_H(G^*)}{\mu_0} \right). \quad (\text{E.69})$$

On the other hand, $\hat{\omega}_\beta^2 = \frac{K}{K-1} \left[\frac{1}{K} \sum_{k=1}^K (\beta_k^*)^2 - (\bar{\beta}^*)^2 \right]$. The SLLN gives $\frac{1}{K} \sum_{k=1}^K (\beta_k^*)^2 \rightarrow \mathbb{E}[(\beta^*)^2]$ and $\bar{\beta}^* \rightarrow \mathbb{E}[\beta^*]$. Therefore, by the continuous mapping theorem, we have:

$$(\bar{\beta}^*)^2 \xrightarrow{a.s.} (\mathbb{E}[\beta^*])^2 \quad (\text{E.70})$$

Since $K/(K-1) \rightarrow 1$ as $K \rightarrow \infty$, thus

$$\hat{\omega}_\beta^2 \rightarrow \mathbb{E}[(\beta^*)^2] - (\mathbb{E}[\beta^*])^2 = \mathbb{V}[\beta^*] = \omega_\beta^2 \quad (\text{E.71})$$

Q. E. D. □

Proposition 9 (Asymptotic normality). Under the normal-approximation assumption for β^* ,⁴ construct $Z_\beta^* = \left[s_\theta^2 (B\hat{\theta}^2)^{-1} + s_\mu^2 (R\hat{\mu}_0^2)^{-1} + \hat{\omega}_\beta^2 \right]^{-1/2} \log \left(\frac{\hat{\theta}}{\hat{\mu}_0} \right)$ by employing $\hat{\omega}_\beta^2$ as the plug-in estimator. Then, $Z_\beta^* \overset{\text{approx}}{\sim} \mathcal{N}(0, 1)$.

⁴Since the targeted subgraph is not too rare in random networks, its distribution may be assumed to be closer to normal distribution.

Proof. The normal-approximation assumption for β^* implies $\beta^* \sim N(0, \omega_\beta^2)$. The null hypothesis requires that the β of the target network converges in distribution to the β^* of the random network, i.e., $\beta \xrightarrow{d} \beta^* \sim N(0, \omega_\beta^2)$. Let $\hat{\beta} = \beta + \varepsilon_\beta$, then for $\mathbb{V}[\hat{\beta}] = \mathbb{V}[\beta] + \omega_\beta^2$, we have $\hat{\beta} \sim N(0, \mathbb{V}_{\beta, MC} + \omega_\beta^2)$, where $\mathbb{V}_{\beta, MC} = \frac{\mathbb{V}_\theta}{\theta^2} + \frac{\mathbb{V}_\mu}{\mu_0^2}$ is the Monte Carlo variance of β .

Now employ $\hat{\mathbb{V}}_{\beta, MC} = \frac{s_\theta^2}{B\hat{\theta}^2} + \frac{s_\mu^2}{R\hat{\mu}_0^2}$ and $\hat{\omega}_\beta^2 = \frac{1}{K-1} \sum_{k=1}^K \left(\hat{\beta}_k^* - \bar{\hat{\beta}}^* \right)^2$ as plug-in estimators of $\mathbb{V}_{\beta, MC}$ and ω_β^2 , respectively. Since $\hat{\mathbb{V}}_{\beta, MC} \rightarrow \mathbb{V}_{\beta, MC}$ and $\hat{\omega}_\beta^2 \rightarrow \omega_\beta^2$, Slutsky's theorem gives:

$$Z_\beta^* = \frac{\hat{\beta}}{\sqrt{\frac{s_\theta^2}{B\hat{\theta}^2} + \frac{s_\mu^2}{R\hat{\mu}_0^2} + \hat{\omega}_\beta^2}} \xrightarrow{d} N(0, 1) \quad (\text{E.72})$$

Q. E. D. □

We further consider relaxing the distributional assumptions. Since the null distribution of sub-graph counts is skewed, heavy-tailed, discrete, zero-inflated, or in cases of cliques, bicliques, dense induced subgraphs, or highly overlapping subgraphs are considered, the null distribution will not be well approximated by a normal distribution. In such cases, the statistic Z_β^* may yield p -values that are either too optimistic or too conservative. Thus, here, consider an empirically calibrated p -value,

which allows us to estimate the probability that the subgraph enrichment coefficient based on a random network is at least as large as that in the given network, i.e., $\mathbb{P}(\beta^* \geq \beta)$. For the null hypothesis $H_0 : G \sim \mathcal{P}_0$ and the right-tail alternative $H_1 : N_H(G) > \mathbb{E}[N_H(G^*)]$, the empirically calibrated p -value for the right-tail test is constructed as $\hat{p}_{\text{cal}}^+ = K + 1^{-1} \left[1 + \sum_{k=1}^K \mathbb{1}(\hat{\beta}_k^* \geq \hat{\beta}) \right]$. Here we have the following propositions:

Proposition 10 (Consistency). $\hat{p}_{\text{cal}}^+$ is a consistent estimator of $\mathbb{P}(\beta^* \geq \beta)$.

Proof. As shown above, we have almost sure convergence conditions including $\hat{\theta} \rightarrow \theta$, $\hat{\mu}_0 \rightarrow \mu_0$, and $\hat{\theta}_k^* \rightarrow \theta_k^* = N_H(G_k^*)$ for each random network. Since $\theta > 0$, $\mu_0 > 0$, and $\theta_k^* > 0$ the Continuous Mapping Theorem gives: $\hat{\beta} \rightarrow \beta = \log \left(\frac{\theta}{\mu_0} \right)$ and $\hat{\beta}_k^* \rightarrow \beta_k^* = \log \left(\frac{\theta_k^*}{\mu_0} \right)$. Therefore,

$$\hat{p}_{\text{cal}}^+ \rightarrow p_{K,\text{cal}}^+ = \frac{1 + \sum_{k=1}^K \mathbb{1}(\beta_k^* \geq \beta_{\text{obs}})}{K + 1} \quad (\text{E.73})$$

Furthermore, as $K \rightarrow \infty$, the Glivenko–Cantelli theorem gives:

$$\hat{p}_{\text{cal}}^+ \rightarrow p_{\text{cal}}^+ = \mathbb{P}(\beta^* \geq \beta) \quad (\text{E.74})$$

Q. E. D. □

Proposition II (Validity). $\widehat{p}_{\text{cal}}^+$ is a valid estimator of $\mathbb{P}(\beta^* \geq \beta)$.

Proof. For a continuous distribution, the probability of any two statistics in the collection $T = \{\beta, \beta_1^*, \beta_2^*, \dots, \beta_K^*\}$ being equal is 0, i.e., $\Pr(T_i = T_j) = 0$, $\forall i \neq j$. This means that the statistics in T can almost surely be strictly ordered, i.e., $T_{(1)} < T_{(2)} < \dots < T_{(K+1)}$. Consequently, the rank of the observed statistic β in the collection $\{\beta, \beta_1^*, \beta_2^*, \dots, \beta_K^*\}$ is uniformly distributed.

Denote $R = 1 + \sum_{k=1}^K \mathbb{1}(\beta_k^* \leq \beta)$ as the rank of β in increasing order. Since the $K+1$ statistics are i.i.d. and have no ties, R follows a discrete uniform distribution, i.e., $R \sim \text{Uniform}\{1, 2, \dots, K+1\}$. According to the construction, p_{cal}^+ employs the number of statistics greater than or equal to β . Hence, since there are no ties, $\sum_{k=1}^K \mathbb{1}(T_k^* \geq T_{\text{obs}}) = K+1 - R$, and therefore $p_{\text{cal}}^+ = \frac{K+2-R}{K+1}$. Meanwhile, since R is uniformly distributed on $\{1, \dots, K+1\}$, p_{cal}^+ is uniformly distributed on the discrete set $\{\frac{1}{K+1}, \frac{2}{K+1}, \dots, 1\}$. Thus, for any α ,

$$\mathbb{P}(p_{\text{cal}}^+ \leq \alpha) = \frac{\lfloor (K+1)\alpha \rfloor}{K+1} \leq \alpha \quad (\text{E.75})$$

Q. E. D.

□

Appendix F Application in International Trade

Here, I report robustness checks for the motif verification of international trade networks, which are mentioned but not presented in detail in the main text. As noted in the main text, the simple random networks constructed for motif verification are based on RCA, and different RCA thresholds are imposed to distinguish three cases, thereby examining the dominant structure and the residual structure.

Accordingly, the robustness checks here are conducted by changing these thresholds. The first check examines the global structure by imposing both lower and upper bounds, namely, $0.1 < RCA \leq 10$. At the same time, I examine the dominant structure using the threshold $1 < RCA \leq 10$, and the residual structure using the threshold $0.1 < RCA \leq 1$. The second check also imposes both lower and upper bounds, but further narrows the range. Specifically, I examine the global structure using the threshold $0.2 < RCA \leq 5$, and examine the dominant and residual structures using the thresholds $1 < RCA \leq 5$ and $0.2 < RCA \leq 1$, respectively. In addition, I replace the Erdős–Rényi null model with the Chung–Lu null model for an additional robustness check, while using the same RCA thresholds as in the main text. The results are shown in Figs.F.4, F.5, and F.6.

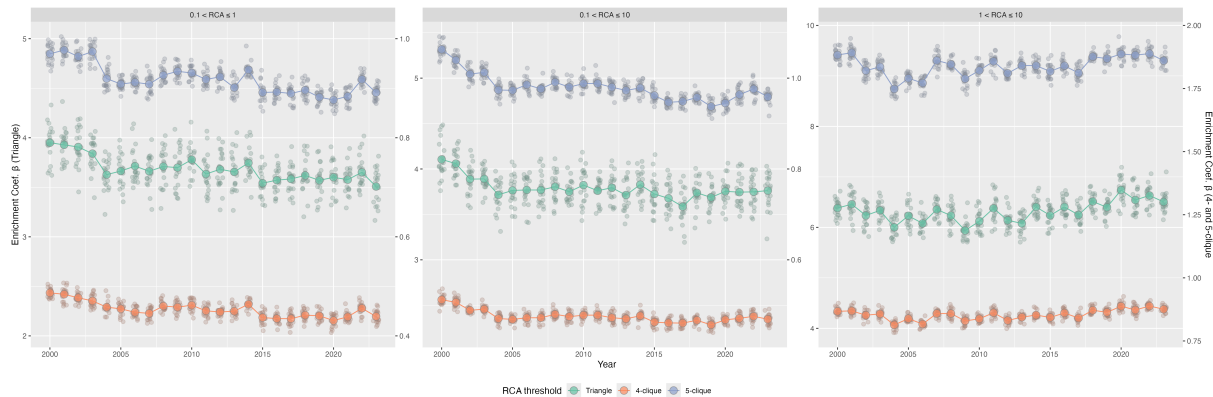


Figure F.4: Motif verification of international trade from 2000 to 2023 ($0.1 < RCA \leq 10$).

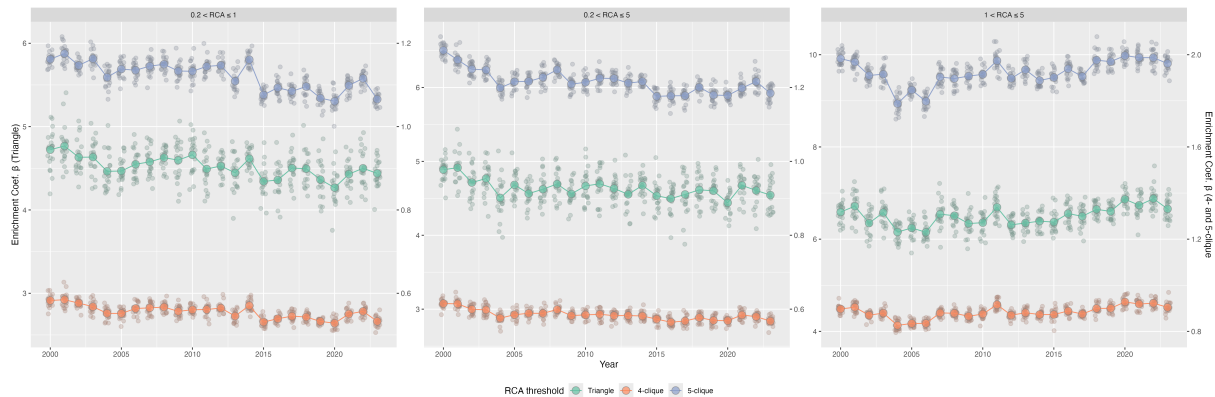


Figure F.5: Motif verification of international trade from 2000 to 2023 ($0.2 < RCA \leq 5$).

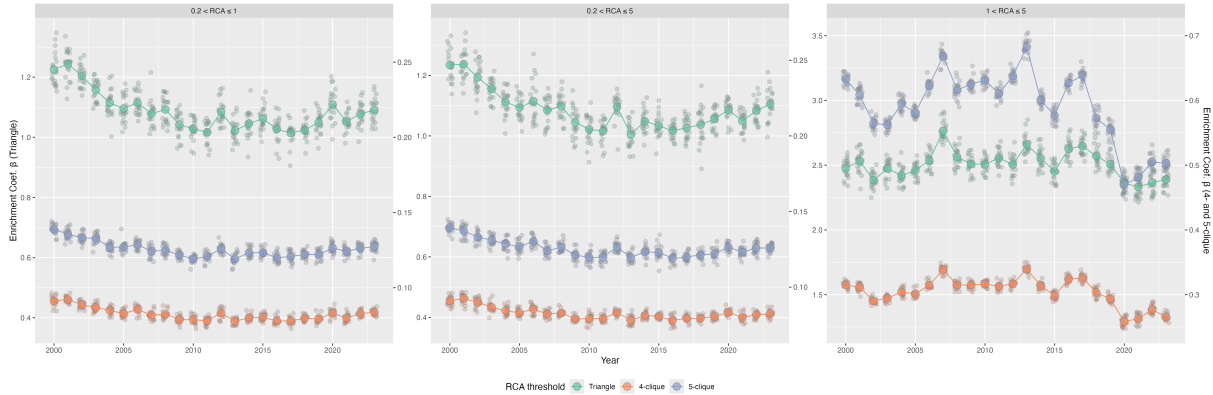


Figure F.6: Motif verification of international trade from 2000 to 2023 (Chung-Lu null model).

The results of both checks are broadly consistent with those in the main text—for the global structure, the enrichment coefficients of the three target motifs all exhibit a declining trend over the past two decades, and such a decline is mainly driven by the residual structure.